Bayesian Structural Inference for Dynamic Crowdsourcing Contests

Linsheng Zhuang
Institute of Operations Research and Analytics
National University of Singapore, Singapore
Center for Intelligent Computing
Great Bay University, China
zhuangls@gbu.edu.cn

Jussi Keppo
NUS Business School & Institute of Operations Research and Analytics
National University of Singapore, Singapore
keppo@nus.edu.sg

Zhi Chen
NUS Business School
National University of Singapore, Singapore
bizczhi@nus.edu.sg

Online crowdsourcing contests generate rich dynamic behavior, but their core strategic state is latent: contestants react to noisy public leaderboard feedback rather than true relative performance. We develop a tractable dynamic game with Bayesian filtering and a corresponding structural estimation framework to recover effort costs, innovation risk, and feedback precision. We prove that a finite-precision design can strictly outperform every full-disclosure design when the prize, deadline, and feedback precision are jointly chosen, even without a direct precision cost. Moreover, the model admits a closed-form leading-order effort policy in which signal precision determines how widely prize incentives extend across public performance gaps, enabling tractable Bayesian estimation. For validation, we use a path-wise holdout design: private-leaderboard path values are excluded from the estimation likelihood and subsequently used for diagnostic tests, although final private ranks are used to define the focal rivalry. In 73 Kaggle contests, higher estimated innovation risk is associated with greater private-leaderboard volatility, whereas higher estimated feedback precision is associated with less public–private disagreement. Both relationships are also supported by time-split tests that compare parameters estimated from early public contest data with subsequent private-leaderboard outcomes.

Key words: Bayesian Statistics, Structural Estimation, Crowdsourcing Contest, Dynamic Contest

1. Introduction

Crowdsourcing contests are widely used by firms to source solutions to complex problems from external contributors. Platforms such as Kaggle[1], Topcoder[2], DrivenData[3], and AIcrowd[4] host thousands of these contests, attracting data scientists, engineers, and developers to compete by submitting solutions to real-world problems. On Kaggle, for instance, a typical contest spans several weeks, during which participants compete for monetary prizes by repeatedly submitting solutions, observing their relative ranking on a public leaderboard, and adjusting their strategies in response to evolving feedback and the observed actions of competitors.

The central empirical challenge in these online crowdsourcing contests is that the true competitive state is unobserved during the contest. Participants observe noisy public leaderboard updates rather than true relative performance, and they choose effort based on filtered beliefs rather than on the latent state itself. The econometrician faces an analogous partial-observability problem during estimation, making it difficult to recover strategic behavior and incentives from contest-phase data alone.

Motivated by this identification problem, we develop a dynamic contest model and structural Bayesian estimation framework tailored to settings such as Kaggle, where contestants choose continuous effort while learning from noisy public leaderboard signals. Applying the framework to the Meta-Kaggle dataset (Risdal and Bozsolik 2022), we recover latent competition states and structural primitives from contest-phase public information and evaluate them using post-contest private-leaderboard paths excluded from the likelihood.

To illustrate the framework in greater detail, Figure 1 depicts the key components and their interactions in a typical Kaggle contest. One notable feature is the existence of two leaderboards: a *private* leaderboard, which is hidden until the contest ends and ultimately determines the final ranking and prize allocation, and a *public* leaderboard, which is visible to all participants during the contest.[5] Leaderboard displays are contestant-specific and update asynchronously. After observing the public leaderboard, contestants update their beliefs and form a perceived state of the competition. Based on these beliefs, contestants choose effort according to the model's Markov perfect equilibrium (MPE) strategies. These effort levels shape submission intensity. Each submission generates a new score for the submitting contestant, while the displayed score of the opponent

[1] Kaggle: https://www.kaggle.com

[2] Topcoder: https://www.topcoder.com

[3] DrivenData: https://www.drivendata.org

[4] AIcrowd: https://www.aicrowd.com

[5] This design serves two purposes. First, the public leaderboard provides participants with real-time feedback. Second, by keeping the private leaderboard concealed, the platform discourages overfitting to the public metric and preserves uncertainty, thereby sustaining competitive tension throughout the contest.

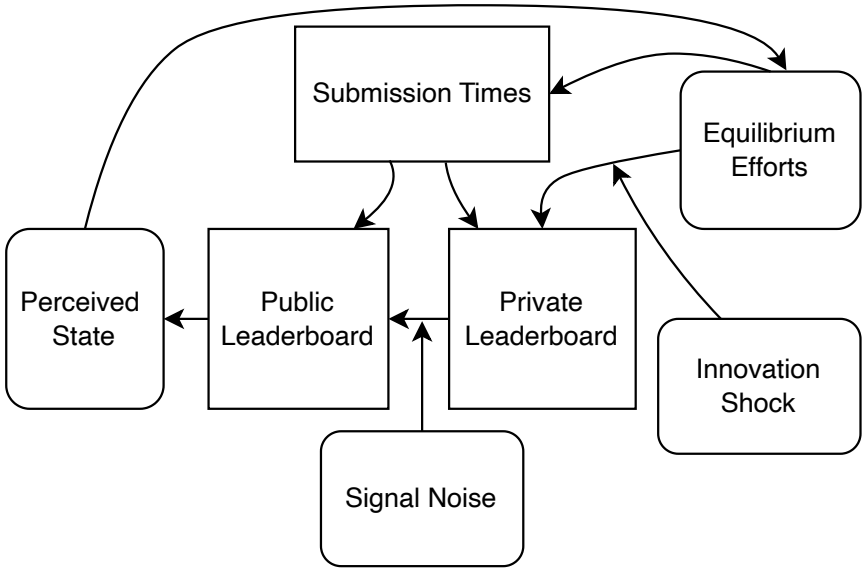


Figure 1 Structure of a Typical Kaggle Contest

Note: This figure illustrates the structure of a typical Kaggle contest and the information flow available to contestants. Rectangles and rounded rectangles represent observed data and latent states, respectively. The private leaderboard is visible only to the contest organizer during the contest and is revealed to all participants only after it ends. During the contest period, contestants can observe only the submission times of each player and the public leaderboard.

remains at its most recent submitted value. In the model, latent performance evolves according to the real-time effort level and an innovation shock.

We estimate the structural parameters of the model using a Bayesian approach, based on contestants' submission times and the observed *public* leaderboard trajectories from the Meta-Kaggle database. This procedure recovers latent competition states, effort dynamics, participant cost parameters, innovation uncertainty, and the precision of the public measurement process.

To assess whether the estimated model captures meaningful contest primitives, we use private-leaderboard paths to evaluate two predictions illustrated in Figure 1: higher estimated innovation risk should be associated with greater private-path volatility, whereas higher estimated signal precision should be associated with smaller public-private leaderboard gaps. Although private-leaderboard path values are excluded from estimation, final private ranks are used to select the focal pair. We therefore interpret the validation exercises as path-wise holdout diagnostics conditional on ex post pair selection. To strengthen the credibility of these diagnostics, we also conduct time-split exercises that relate primitives estimated from early public contest data to private-path outcomes in the later contest period, thereby reducing the concern that the full-path relationships merely reflect common path-level variation.

Empirically, we carefully select 73 high-quality contests from the Meta-Kaggle database and preprocess the data to ensure consistency across estimation methods. Conditional on the focal-pair selection rule, we estimate contest-specific innovation risk and signal precision using only public contest-phase information. The validation results show that structurally estimated innovation risk is positively associated with private-leaderboard volatility, including in a time-split diagnostic. Higher estimated signal precision is associated with smaller public-private leaderboard disagreement, including in the time-split diagnostic, and the positive relationship between structural and benchmark precision estimates also appears under an alternative focal-pair specification. Using a steady-state, leading-order calibration at the observed prize and duration, we then derive contest-specific precision recommendations and quantify their expected output gains.

This paper makes three main contributions. First, we develop a dynamic game-theoretic model and a Bayesian structural framework for contests with noisy public feedback and latent performance dynamics. Relative to contest models centered on discrete submit-or-stop decisions, our approach models continuous effort accumulation and makes belief updating from noisy signals structurally central. Building on the equilibrium structure of Ryvkin (2022), we model the winner as determined by the private leaderboard, hidden until the contest ends, so that the informativeness of public feedback shapes equilibrium effort incentives, while retaining a closed-form leading-order effort policy for structural estimation. Second, we establish conditions under which a feasible finite-precision design strictly outperforms every full-disclosure design, even without a direct precision cost. Consequently, full disclosure is not jointly optimal in that region, and any attained joint optimum must use finite precision. Third, we use private-leaderboard path values, which are excluded from the likelihood, to construct post-estimation validation targets. This approach allows us to assess whether the estimated structure captures innovation uncertainty and signal precision rather than merely fitting observed contest-phase behavior.

1.1. Related Literature

This study is most closely related to the following strands of literature.

Contest Design. A rich literature on *static* contest design (Ales et al. 2017) investigates how to structure competitive environments to elicit performance, typically through a contest success function mapping effort to winning probabilities (Skaperdas 1996).[6] Because our setting is dynamic with repeated feedback, this static branch serves mainly as background for the mechanism-design questions we build on, including prize allocation, participation rules, and information disclosure (Moldovanu and Sela 2001, Ales et al. 2021, Bergemann et al. 2022).

[6] There are exceptions, such as beauty or barrier contests.

While these studies offer valuable insights into incentive design, they largely overlook how participants dynamically adjust their efforts in response to evolving competition. In contrast, this paper focuses on *dynamic* contests, where strategic decisions unfold over time. Two main modeling approaches have been used to capture such dynamic strategic interactions:

One strand of the literature models dynamic contests as multistage processes, where participants exert effort over time while facing uncertainty about intermediate progress. This structure facilitates the analysis of the mid-contest policy instruments available to the contest designer, such as feedback disclosure and screening mechanisms. Specifically, in a typical two-stage setting (Aoyagi 2010, Mihm and Schlapp 2019), the initial stage captures either feasibility uncertainty or foundational experimentation, with progress revealed through milestone feedback provided by the contest organizer. Participants exert effort in both stages, and the final output reflects the cumulative impact of early performance and subsequent improvement. Similarly, Bimpikis et al. (2019) study a two-stage model in which agents exert continuous effort to complete Stage A (potentially infeasible) and Stage B (feasible), with breakthroughs governed by Poisson processes. In all these models, the contest designer may strategically disclose interim feedback. Another example is Khorasani et al. (2023), who analyze a two-stage contest in which solvers invest exploratory effort to generate viable ideas, with only those passing an imperfect screening mechanism proceeding to execution.

The second approach, more closely related to our work, treats output as a stochastic process whose drift depends on the participant's effort. This line of research originates from the tug-of-war framework, first formalized by Harris and Vickers (1987) as a one-dimensional simplification of a multi-stage R&D race. Subsequent studies build on this foundation by modeling each player's progress as a Brownian motion. Budd et al. (1993), for example, analyze a duopoly innovation race where the state evolves as a Brownian motion driven by the difference in effort, and characterize an approximate equilibrium. Moscarini and Smith (2011) advance this approach by modeling the game state as the output gap between players and derive closed-form equilibrium strategies under pure-strategy play. Ryvkin (2022) adapt their framework to a dynamic contest with a fixed deadline, providing a closed-form solution for the Markov perfect equilibrium.

The precision of public feedback is itself a design instrument in data-science contests. Huang et al. (2024) study the training–test split, showing that more informative interim feedback makes the ranking statistically more reliable but also gives contestants the strongest incentive to submit high-variance, suboptimal models; evidence from recent Kaggle contests is consistent with this incentive effect. In contrast, our model features a different channel: more precise signals improve belief filtering and make the public ranking a more reliable predictor of the private winner.

This paper studies online crowdsourcing contests, which typically feature fixed deadlines, repeated submissions, and public real-time feedback, all governed by platform-specific rules and

evaluation metrics. Hence, our analysis builds on the model by Ryvkin (2022), whose simplifications yield a closed-form equilibrium that makes structural estimation in a dynamic setting analytically and computationally tractable. We depart from their full-information setting by introducing a noisy public signal and Bayesian filtering, thereby bringing partial observability into a tractable dynamic contest model.

Structural Estimation of Contests. Beyond theoretical modeling, a growing body of research employs structural estimation techniques to recover key parameters of contest environments using observed data.

In the structural estimation of static contests, when a contest success function (CSF) is explicitly specified, the primary goal is to estimate its functional form or key parameters (Hwang 2012, Kang 2016, Huang and He 2021). In contrast, for contests without a well-defined or smooth CSF, such as beauty contests, researchers typically rely on equilibrium conditions, most notably the first-order condition. For example, Yoganarasimhan (2016) proposes a two-step semiparametric estimator, resembling the EM algorithm, to recover both unobserved auction heterogeneity and bidder cost distributions. Similarly, in auction-like models (Guerre et al. 2000, Shakhgildyan 2022), nonparametric estimation of the underlying valuation structure often serves as a first step, providing the foundation for subsequent structural estimation.

Most dynamic structural estimation exercises rely on the Markov Perfect Equilibrium (MPE) of a dynamic game. However, when the model is calibrated to closely fit real-world data, the MPE typically lacks a closed-form solution. For example, Jiang et al. (2022) develop a finite-horizon dynamic game with discrete time, state, and action spaces to model creators' strategic behavior in response to feedback during online logo design contests. Using data from 810 contests, they implement a two-step maximum likelihood estimation (MLE) procedure under the assumption of ex-ante homogeneous contests and players: the first step estimates action-specific state transition probabilities, while the second recovers utility parameters.

Specific to online crowdsourcing environments, Zhang et al. (2019) build a dynamic structural model to study how participants in crowdsourcing contests learn and choose contests over time. Their model shows that competing against superstars accelerates learning by providing more informative feedback. Using Bayesian updating and a hierarchical model, they estimate how individuals trade off monetary rewards, learning opportunities, and participation costs, based on data from Topcoder.

Most closely related to our work, Lemus and Marshall (2021) structurally estimate contest behavior using Kaggle data. They propose a finite-horizon dynamic game in which, after each submission, players decide whether to continue or exit, essentially solving an optimal stopping problem. Players can only make this decision immediately after completing a submission, which arrives at random

intervals and incurs a privately drawn cost. Their estimation proceeds in two stages: the first uses maximum likelihood and the EM algorithm to estimate primitive parameters such as the distribution of player types and scores; the second applies a conditional choice probability (CCP)-based estimator to identify the cost distribution. Our model differs in two key respects: (1) we link submission intensity to current effort rather than model submissions as discrete costly actions; and (2) we explicitly model belief updating from noisy signals as a structural channel that shapes both behavior and identification, whereas their CCP-based approach relies on a coarser behavioral approximation.

The structure of the paper is as follows: Section 2 introduces our theoretical model of online crowdsourcing contests. Section 3 presents the Bayesian estimation framework. In Section 4, we describe the data-generating process and apply the Bayesian framework to synthetic data. Section 5 applies the framework to real-world Kaggle data and evaluates the estimated primitives using private-leaderboard path values excluded from the likelihood. A replication package (code and implementation details) will be provided with the paper.

2. The Model

Two players, i and j, compete for a prize $\theta > 0$ in a contest. The winner receives the prize and the loser receives zero. Following Ryvkin (2022), the contest starts at time zero and ends at a fixed deadline $T > 0$. At each $t \in [0, T)$, each player $k \in \{i, j\}$ chooses effort $q_{k,t} \geq 0$ and incurs an instantaneous quadratic cost $C_k(q_{k,t}) = c_k q_{k,t}^2/2$, where $c_k > 0$ and a lower c_k corresponds to higher ability. Let y_t denote the *latent private-score gap* between players i and j at time t, evolving as

$$dy_t = (q_{i,t} - q_{j,t})dt + \sigma dW_t \tag{1}$$

where W_t is a Brownian motion and $\sigma > 0$ measures the innovation risk. For given efforts, a larger σ increases the uncertainty in performance changes.

The actual contest is equipped with a submission system in which contestants submit solutions asynchronously over time. Each submission generates new public and private scores for the submitting contestant, while the opponent's most recently submitted scores are carried forward when constructing the corresponding leaderboard gaps. For tractability, we approximate this irregular sequence of submission-triggered updates by a continuous-time public signal of the latent private-score gap y_t. The feasible precision parameter satisfies $\lambda > 0$, and the signal evolves by

$$dZ_t = y_t dt + \frac{dB_t}{\sqrt{\lambda}} \tag{2}$$

where B_t is a standard Brownian motion independent of W_t. A larger λ implies less noise and more informative feedback; in the empirical mapping, this corresponds to a more informative public

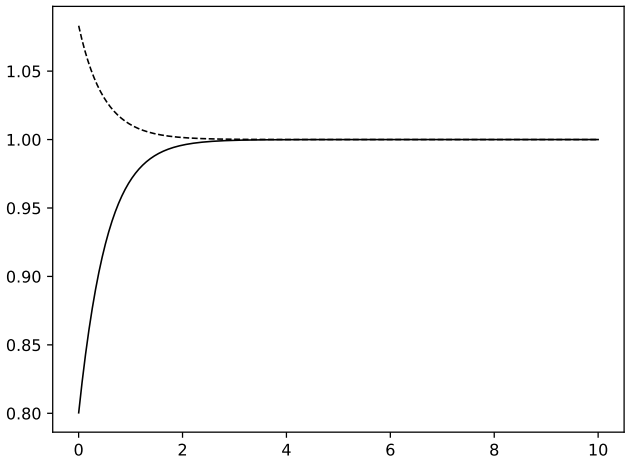


Figure 2 Convergence of the posterior variance S_t to steady state.

Note: With $\lambda = \sigma = 1$, the solid curve starts below the steady state ($S_0 < 1$) and the dashed curve starts above it ($S_0 > 1$). Both converge to $\bar{S} = 1$. The horizontal axis is time t; the vertical axis is posterior variance S_t.

leaderboard signal about latent relative performance. This continuous-time signal is an idealized analogue of the stream of public leaderboard updates observed after submissions.

The model parameters are common knowledge. Public submission activity provides a proxy for effort; the theoretical model idealizes this indirect observation by treating past efforts as observable to both players. Neither player has additional private information about y_t. We study public Markov equilibria. Let I_t denote the common history of public signals and past efforts, and define the posterior mean and variance by $\tilde{y}_t := E(y_t \mid I_t)$ and $S_t := E[(y_t - \tilde{y}_t)^2 \mid I_t]$. Initially, $y_0 \mid I_0 \sim \mathcal{N}(\tilde{y}_0, S_0)$, where $\tilde{y}_0$ is publicly known and $S_0 > 0$ is fixed independently of λ. The initial filtering error is independent of future increments of W and B. Under standard linear-Gaussian assumptions, the Kalman-Bucy filter (Chapter 1.2 of Bensoussan 1992; see also Frogerais et al. 2012, Barrau and Bonnabel 2017) implies

$$d\tilde{y}_t = (q_{i,t} - q_{j,t})dt + \sqrt{\lambda} S_t \, d\widehat{B}_t \tag{3}$$

$$\frac{dS_t}{dt} = \sigma^2 - \lambda S_t^2 \tag{4}$$

$d\widehat{B}_t := \sqrt{\lambda}(dZ_t - \tilde{y}_t dt)$ is the innovation Brownian motion, so equation (3) is equivalent to the usual Kalman–Bucy representation. Hence, the conditional distribution satisfies $y_t \mid I_t \sim \mathcal{N}(\tilde{y}_t, S_t)$, so the posterior state is fully characterized by $(\tilde{y}_t, S_t)$. The steady-state value associated with equation (4) is $\bar{S} \equiv \sigma/\sqrt{\lambda}$. Given S_0, equation (4) determines a deterministic path S_t, so $(\tilde{y}_t, t)$ remains a sufficient strategic state. Appendix B.3 reports the solution.

The sign of y_T determines the winner. Let I_T denote the common information just before the terminal private scores are revealed. Since $y_T \mid I_T \sim \mathcal{N}(\tilde{y}_T, S_T)$, the expected terminal prizes are

$G_i(\tilde{y}_T) = \theta\Phi(\tilde{y}_T/\sqrt{S_T})$ and $G_j(\tilde{y}_T) = \theta\Phi(-\tilde{y}_T/\sqrt{S_T})$, where Φ is the standard normal cumulative distribution function. Let $m_k(\tilde{y},t)$ denote player k's equilibrium Markov effort strategy. Taking player j's strategy $q_{j,\tau} = m_j(\tilde{y}_\tau,\tau)$ as given, player i's value is

$$V^i(\tilde{y},t) = \sup_{\{q_{i,\tau}\geq 0\}_{\tau\in[t,T)}} \mathbb{E}\left[G_i(\tilde{y}_T) - \int_t^T C_i(q_{i,\tau})d\tau \,\middle|\, \tilde{y}_t = \tilde{y}\right],$$

The supremum is over nonnegative effort processes predictable with respect to I_τ and satisfying $\mathbb{E}\int_t^T q_{i,\tau}^2 d\tau < \infty$. The optimization is subject to equations (3)–(4). Player j's problem is obtained by replacing $\tilde{y}$ with $-\tilde{y}$ and exchanging the player labels and cost parameters. The corresponding Hamilton-Jacobi-Bellman (HJB) equation for player i is

$$0 = \max_{q_{i,t}\geq 0}\left[-\frac{c_i q_{i,t}^2}{2} + V_y^i\cdot(q_{i,t} - m_j(\tilde{y},t)) + V_t^i + \frac{V_{yy}^i}{2}\lambda S_t^2\right].$$

Here, subscripts y denote derivatives with respect to the posterior mean $\tilde{y}$. For an interior equilibrium, substituting the first-order conditions $q_{i,t} = V_y^i/c_i$ and $q_{j,t} = -V_y^j/c_j$ yields the system

$$\frac{1}{2c_i}(V_y^i)^2 + \frac{1}{c_j}V_y^iV_y^j + V_t^i + V_{yy}^i\frac{\lambda S_t^2}{2} = 0$$
$$\frac{1}{2c_j}(V_y^j)^2 + \frac{1}{c_i}V_y^jV_y^i + V_t^j + V_{yy}^j\frac{\lambda S_t^2}{2} = 0$$

subject to boundary conditions $V^i(-\infty,t) = 0$, $V^i(+\infty,t) = \theta$, $V^j(-\infty,t) = \theta$, $V^j(+\infty,t) = 0$, and terminal conditions $V^i(\tilde{y},T) = G_i(\tilde{y})$ and $V^j(\tilde{y},T) = G_j(\tilde{y})$ for all $\tilde{y}\in\mathbb{R}$. The following theorem establishes equilibrium existence and gives a leading-order effort formula for small incentive indices.

THEOREM 1 **(Equilibrium Existence and Leading-Order Effort)**. *For every finite $\lambda > 0$ and sufficiently small $w_i + w_j$, the game admits an interior Markov-perfect equilibrium with bounded, smooth effort policies. As $(w_i,w_j)\to(0,0)$, the effort policies of this equilibrium satisfy*

$$m_k(\tilde{y}_t,t) = \frac{\sigma^2 w_k}{\sqrt{Q_t}}\left\{\frac{1}{\sqrt{2\pi}}\exp\left[-\frac{\tilde{y}_t^2}{2Q_t}\right] + O(w_i+w_j)\right\}, \tag{5}$$

for $k\in\{i,j\}$, where $w_k := \theta/(\sigma^2 c_k)$ and $Q_t := S_t + \sigma^2(T-t)$. The remainder is uniform over $(\tilde{y}_t,t)\in\mathbb{R}\times[0,T)$, holding T,σ,λ, and S_0 fixed.

The quantity Q_t combines current uncertainty about the true gap, S_t, with innovation risk over the remaining horizon, $\sigma^2(T-t)$. To leading order, effort is proportional to θ/c_k and is highest when the common posterior gap is zero, where a marginal improvement most strongly affects the probability of winning. A larger Q_t lowers this peak but sustains effort over a wider range of gaps.

2.1. Joint Design of Prize, Deadline, and Feedback Precision

The performance gap is sufficient for the contestants' strategic problem, but the organizer values the best terminal performance. We therefore recover individual private performance scores. Let x_t^k denote contestant k's score, normalized so that $(x_0^i + x_0^j)/2 = 0$. The initial gap is $y_0 := x_0^i - x_0^j$, with $y_0 \mid I_0 \sim \mathcal{N}(\tilde{y}_0, S_0)$. Individual scores evolve according to

$$dx_t^k = q_{k,t}dt + \frac{\sigma}{\sqrt{2}}dW_t^k, \qquad k \in \{i, j\}, \tag{6}$$

where W^i and W^j are independent and $q_{k,t} = m_k(\tilde{y}_t, t)$ in equilibrium. With $W_t = (W_t^i - W_t^j)/\sqrt{2}$, the gap obeys equation (1). The organizer values the best terminal score,

$$\mathcal{B}(\tilde{y}_0, \theta, T, \lambda; S_0) := \mathbb{E}[\max\{x_T^i, x_T^j\} \mid I_0].$$

The organizer chooses the prize, deadline, and public-signal precision simultaneously. Hence $\mathcal{B}(0, 0, T, \lambda; S_0) = \sqrt{\frac{S_0 + \sigma^2 T}{2\pi}}$ and $\mathcal{B}(0, \theta, 0, \lambda; S_0) = \sqrt{\frac{S_0}{2\pi}}$.

With the symmetric initial belief $\tilde{y}_0 = 0$, the organizer jointly chooses

$$\sup_{\theta, T \geq 0,\ \lambda \in (0,\infty]} \{u(\mathcal{B}(0, \theta, T, \lambda; S_0)) - \theta - D(T)\}. \tag{7}$$

Here $u : [0, \infty) \to \mathbb{R}$ is twice continuously differentiable, strictly increasing, and concave. The delay cost $D : [0, \infty) \to [0, \infty)$ is continuously differentiable, strictly increasing, and convex, with $D(0) = D'(0) = 0$ and $D'(T)/T$ nondecreasing for $T > 0$. It is twice continuously differentiable near zero, with $0 < D''(0) < \infty$. Short delays therefore have a locally quadratic cost. The design set requires a positive amount of public feedback. The endpoint $\lambda = \infty$ denotes the full-disclosure game: contestants observe the initial gap and its subsequent path, and expected output averages over the initial prior. For this benchmark, we use the symmetric Markov equilibrium constructed in the appendix.

PROPOSITION 1 **(Feedback Precision in Joint Design)**. *Suppose $c_i = c_j = c > 0$, $-u''(B) \geq \zeta u'(B)$ for all sufficiently large B and some $\zeta > 0$, and S_0 is sufficiently large. There exist thresholds $c_F, \sigma_F > 0$ with the following properties:*

(i) Finite feedback strictly dominates full disclosure. For $0 < c < c_F$ and $0 < \sigma < \sigma_F$, some positive-prize, positive-deadline, finite-precision design strictly outperforms every full-disclosure design. Every joint optimizer satisfies $\theta^* > 0$, $T^* > 0$, and $\lambda^* < \infty$.

(ii) Positive-prize full disclosure. There exist $c > 0$ and $\sigma > \sigma_F$ for which joint optima exist and satisfy $\theta^* > 0$, $T^* > 0$, and $\lambda^* = \infty$.

(iii) Prize exit at high cost. For every $\sigma > 0$ and sufficiently large c, joint optima exist and satisfy $\theta^* = 0$ and $T^* > 0$. Precision is then payoff-irrelevant.

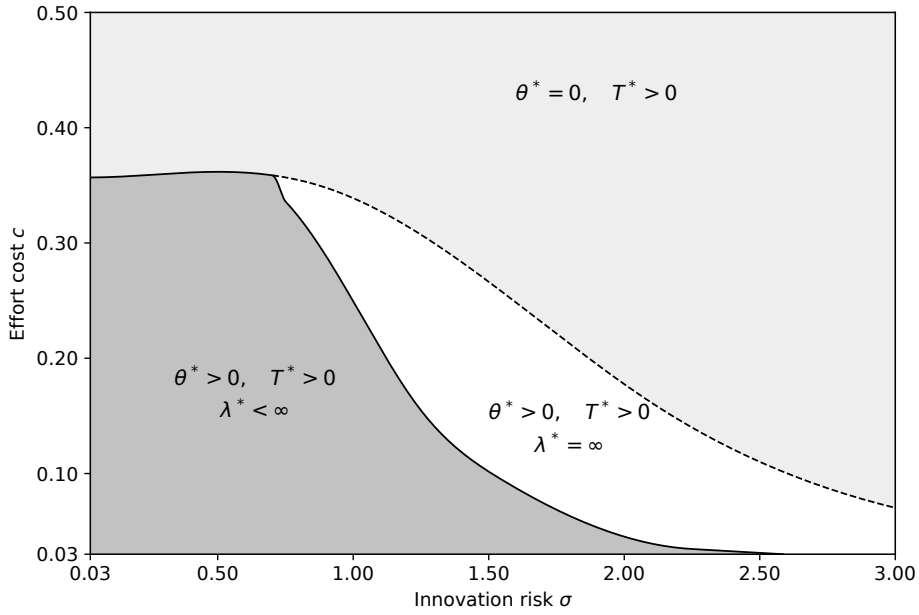


Figure 3 Numerical joint-design regions in the (c, σ) plane

Note: The nonlinear-equilibrium search jointly varies the prize, deadline, and precision, with $S_0 = 1$, $u(B) = 3(1 - e^{-2B})$, and $D(T) = T^2/2$. Medium gray denotes positive prize with finite precision, white positive prize with full disclosure, and light gray zero prize. Solid and dashed lines outline the finite-precision and zero-prize regions, using smoothing-spline fits for display. Both axes use linear scales and start at 0.03.

The proposition identifies sufficient conditions for three design comparisons; Figure 3 illustrates their numerical arrangement in the (c, σ) plane. When effort is inexpensive and innovation risk is low ($c < c_F$, $\sigma < \sigma_F$), retaining some ranking uncertainty can sustain incentives, so finite feedback strictly dominates full disclosure. This comparison does not assert attainment over finite precisions. Full disclosure can nevertheless be jointly optimal at some higher-risk parameter points ($\sigma > \sigma_F$). At sufficiently high effort costs, prize incentives are no longer worthwhile ($\theta^* = 0$), and feedback precision becomes irrelevant.

3. Estimation Framework

This section links equilibrium effort to observed submission timing and public leaderboard updates to estimate the model's structural parameters. We specify the observation process and modeling assumptions, then construct an approximate Bayesian likelihood using the leading-order effort policy in Theorem 1.

3.1. Data-Generating Process

For each Kaggle contest, the observable data from Meta-Kaggle can be grouped into three components.

The first component contains contest-level institutional features: contest duration, prize structure, and information disclosure rules. To match the two-player winner-take-all structure of the model, we restrict the empirical application to contests where a dominant top rivalry is plausible and monetary prize incentives can be mapped to an effective top-two winner-gain structure. The effective prize θ is observed rather than estimated. We express monetary payoffs in units of one hundred thousand U.S. dollars and use this unit for every contest.

The second component records submission events of player i, denoted by $\{\hat{t}_k^i\}_{k=1}^{N_i}$, where N_i is the total number of submissions. We model submissions as counting processes whose intensities depend on equilibrium effort and the preceding public history. The two contestants' submissions are generated using independent Poisson randomness. To link observed submission timing to effort, we assume

$$\tau_i(t) = r \cdot m_i(\tilde{y}_t, t) \tag{8}$$

where $r > 0$ converts model effort into submission intensity; the same relationship applies to player j. This proportional relationship is an empirical measurement assumption rather than a result derived from the theoretical game. In estimation, r is treated as an auxiliary scale parameter rather than a contest primitive.

The third component contains *public* and *private* leaderboard series, denoted by $\hat{x}_t^{i(j)}$ and $x_t^{i(j)}$. In Kaggle contests, public and private scores are computed from different subsets of the test data. Public scores are displayed during the contest, whereas private scores are revealed only after it ends. We define the latent contemporaneous gap as $y_t := x_t^i - x_t^j$ and the displayed public gap as $\hat{y}_t := \hat{x}_t^i - \hat{x}_t^j$, where each displayed score equals the contestant's most recently submitted public score. Because submissions are asynchronous, $\hat{y}_t$ need not compare contemporaneous submissions by the two contestants.

The public gap provides a noisy empirical measure of the latent performance gap. At retained readout time t_k, let $h_k = t_k - t_{k-1}$, with $t_0 = 0$, denote the elapsed time since the preceding retained readout. In the empirical observation model, we assume a fresh Gaussian measurement error $\varepsilon_k \sim \mathcal{N}(0, 1/(\lambda h_k))$. Conditional on the preceding public history and the current update time, this error is independent of the latent state.[7] Substituting the latent state dynamics in equation (1) gives the approximate observation equation

$$\hat{y}_{t_k} \approx y_0 + \int_0^{t_k} [m_i(\tilde{y}_s, s) - m_j(\tilde{y}_s, s)]\, ds + \sigma W_{t_k} + \varepsilon_k. \tag{9}$$

[7] For a fixed interval of length h, equation (2) gives the average readout $(Z_t - Z_{t-h})/h = h^{-1}\int_{t-h}^{t} y_s\, ds + (B_t - B_{t-h})/(\sqrt{\lambda}h)$. The noise term has variance $1/(\lambda h)$. Replacing the interval-average latent gap by its endpoint value motivates the empirical observation model; it is not an exact consequence of sampling the continuous signal at endogenous submission times.

Here, σW_{t_k} captures accumulated innovation noise, while ε_k captures the public-signal noise associated with the leaderboard update. Thus, larger λ implies more precise public feedback.

The realized initial gap is not observed during the contest. Following the initial belief in Section 2, we assume

$$y_0 \sim \mathcal{N}(\mu_0, S_0)\,, \qquad S_0 > 0. \tag{10}$$

We integrate over y_0 in the likelihood. The parameter μ_0 is the initial perceived gap, while S_0 measures uncertainty about the initial gap. Given (S_0, σ, λ), equation (4) determines the deterministic variance path S_t. The precision parameter λ enters both the observation system and the deterministic filtering-variance path.

The theoretical model uses a continuous public signal and therefore admits the continuous-time filtering equation in (3). The empirical leaderboard, however, is observed only at asynchronous submission times. An exact empirical implementation would therefore require a hybrid filter with continuous prediction between submissions and discrete measurement updates.

To retain the continuous-update representation, we carry forward the most recently observed public gap. The structural DGP uses the equilibrium policy m, whereas estimation uses its leading-order approximation $m^{(1)}$, defined in equation (14). Together with the deterministic variance path from equation (4), this gives the empirical state recursion

$$d\tilde{y}_t = \left[m_i^{(1)}(\tilde{y}_t, S_t, t) - m_j^{(1)}(\tilde{y}_t, S_t, t)\right] dt + \lambda S_t(\hat{y}_t - \tilde{y}_t)dt \tag{11}$$

with initial condition $\tilde{y}_0 = \mu_0$. The displayed gap $\hat{y}_t$ is a piecewise-constant empirical path, not a pointwise observation of the theoretical continuous signal Z_t. Between retained updates it is carried forward without a new measurement, so its continuous use may overstate learning. The estimates are therefore conditional on this approximation.

3.2. Modeling Considerations

This subsection discusses three practical modeling concerns that are important for interpretation: strategic submission timing, the two-player approximation, and the normality assumption. The goal is to clarify the scope of the baseline framework and connect each concern to the robustness analyses that follow.

Strategic submission timing. Submission-triggered leaderboard updates create delays in public feedback: no new leaderboard readout arrives during periods without submissions. Strategic timing or withholding can therefore affect the information available to contestants and complicate inference.

Our submission-intensity specification links submissions to effort but does not explicitly model strategic withholding. Strategic submission timing is an important concern in the literature. For

example, Lemus and Marshall (2021) adopt a similar simplification by excluding such behavior from their model. Our estimates should therefore be interpreted for contests where active updating behavior dominates withholding behavior. As a partial mitigation, we require a minimum number of submissions when selecting representative contestants (Section 5.1), which reduces cases where sparse strategic timing dominates observed dynamics.

Selection of Two Players. Another concern is the two-player approximation. Real contests involve many participants, so reducing the environment to two players changes strategic externalities and information structure. Our interpretation is that the model isolates the dominant top rivalry in contests where competition is concentrated near the frontier. In Section 6.1, we conduct a series of robustness checks. By varying the focal-pair definition, we assess whether the validation results depend on a particular choice of representative rivalry.

Gaussian approximation. The model relies on Gaussian signal and innovation shocks. Mapping bounded leaderboard scores into the model's real-valued state space therefore requires both a transformation and a maintained distributional approximation. The empirical transformation is defined in Section 5.1. Rescaling the transformed scores changes their units but does not determine whether their distribution is Gaussian. We therefore treat Gaussianity as a modeling assumption rather than an empirically established property of every contest.

3.3. Bayesian Estimation Framework

We construct an approximate Bayesian posterior for the general parameter vector $\Theta := (c_i, c_j, \sigma, \lambda, \mu_0, r, S_0)$ using the data introduced above. Specific estimation exercises may fix individual parameters or impose restrictions, such as the steady-state initialization $S_0 = \sigma/\sqrt{\lambda}$, and estimate only the remaining free parameters.

Final private ranks are used to select the focal pair. Conditional on this selection, estimation uses the pair's contest-phase submission counts and public leaderboard readouts. The private leaderboard trajectory y_t, observed ex post in Meta-Kaggle, is reserved for post-estimation path validation.

We discretize time into M intervals of length Δ_{num}, with endpoints $s_\ell = \ell\Delta_{\text{num}}$. Let n_ℓ^i denote contestant i's submission count in $(s_{\ell-1}, s_\ell]$, so that $\sum_{\ell=1}^{M} n_\ell^i = N_i$. Holding the submission intensity at its interval-start value gives the submission likelihood contribution (and analogously for j)

$$\mathcal{L}_i(\Theta) \propto \exp\left\{-\Delta_{\text{num}} \sum_{\ell=1}^{M} \tau_i^{(1)}(s_{\ell-1})\right\} \prod_{\ell=1}^{M} \left[\tau_i^{(1)}(s_{\ell-1})\Delta_{\text{num}}\right]^{n_\ell^i}. \tag{12}$$

Here $\tau_i^{(1)}(t) = rm_i^{(1)}(\tilde{y}_t, S_t, t)$ is the leading-order approximation to the submission intensity in equation (8); the omitted count-factorial terms do not depend on Θ. Given the preceding public history, submission counts are generated by independent Poisson draws, independent of the latent initial

gap and innovation shocks. The likelihood includes all submissions, including multiple submissions within an interval; intervals with no submissions contribute through the exponential term.

For the leaderboard likelihood, we retain one public-gap readout at the end of each grid interval containing at least one submission by either contestant. Denote the retained times and gaps by $t_1 < \cdots < t_N$ and $\hat{y}_{t_k}$, respectively, and let $h_k = t_k - t_{k-1}$, with $t_0 = 0$. Conditional on the preceding history, the readout time t_k, and Θ, the empirical state and observation equations imply

$$\hat{y}_{t_k} \mid \mathcal{F}_{k-1}, t_k, \Theta \sim \mathcal{N}\left(\mu_{k|k-1}, V_{k|k-1}\right),$$

Here $\mathcal{F}_{k-1}$ denotes the submission and public-leaderboard history strictly before t_k. The one-step-ahead predictive moments incorporate initial-state uncertainty, accumulated innovation, and preceding readouts. Appendix B.2 gives their explicit sequential recursion. The leaderboard likelihood is therefore

$$\mathcal{L}_{\text{pub}}(\Theta) = \prod_{k=1}^{N} \phi\left(\hat{y}_{t_k} \mid \mu_{k|k-1}, V_{k|k-1}\right). \tag{13}$$

After evaluating each predictive density, we use the readout in the latent-state update and, separately, in the perceived-state recursion in equation (11). The moments $(\mu_{k|k}, P_{k|k})$ belong to the auxiliary filter used for the leaderboard likelihood; the strategic belief pair $(\tilde{y}_t, S_t)$ continues to follow equations (11) and (4). The joint likelihood is proportional to $\mathcal{L}_i(\Theta)\mathcal{L}_j(\Theta)\mathcal{L}_{\text{pub}}(\Theta)$, with all contributions evaluated using the same evolving public history.

We evaluate the likelihood using the leading-order policy from equation (5):

$$m_k^{(1)}(\tilde{y}_t, S_t, t) = \frac{\theta}{c_k\sqrt{2\pi Q_t}} \exp\left(-\frac{\tilde{y}_t^2}{2Q_t}\right), \qquad Q_t = S_t + \sigma^2(T-t). \tag{14}$$

For each parameter vector Θ, equation (4) determines S_t, and equation (11) advances the perceived state and the associated effort paths. The posterior is conditional on both the leading-order policy approximation and the continuous-update approximation.

We combine the joint likelihood with priors on the remaining free parameters to obtain the approximate posterior. Prior distributions, hyperparameters, and numerical parameter bounds are reported with each estimation exercise. We sample the resulting approximate posterior using the No-U-Turn Sampler, an adaptive Hamiltonian Monte Carlo method (Hoffman and Gelman 2014).

Cost asymmetry is necessary for identifying all parameters, as established below.[8]

PROPOSITION 2 **(Local Identification)**. *Fix the initial public input before generating observations, with $y_0 \sim \mathcal{N}(\mu_0, S_0)$ as in equation (10). Suppose $\theta > 0$, the observation grid contains at least three intervals, and $c_i \neq c_j$. Under the maintained leading-order empirical model, the interior parameter vector $\Theta = (c_i, c_j, \sigma, \lambda, \mu_0, r, S_0)$ is locally identified from equations (12) and (13).*

[8] Proposition 2 establishes identification for the fixed-input model, not for the sampling law induced by the data-dependent initialization used in the empirical application; see Section 5.1.

4. Synthetic Experiments

We examine the finite-sample performance of the estimation framework using synthetic contests. Data are generated from the numerically computed equilibrium policy m, while estimation uses its leading-order approximation $m^{(1)}$ in equation (14).

Algorithm 1 Synthetic Data Simulation

Input: $\Delta > 0$, T, θ, c_i, c_j, σ, $\lambda > 0$, r, $\tau^\star$, μ_0, S_0, with $T/\Delta \in \mathbb{N}$

Solve equation (4) for $(S_t)_{t\in[0,T]}$ and the nonlinear HJB system for m_i and m_j.

Draw $y_0 \sim \mathcal{N}(\mu_0, S_0)$.

Sample rate-$\tau^\star$ Poisson candidate sets $\mathcal{C}_i, \mathcal{C}_j$ on $[0,T)$.

Sample $u_s^k \sim \mathcal{U}[0,1]$ for $s \in \mathcal{C}_k$, $k \in \{i,j\}$.

Sample $\Delta W_t \sim \mathcal{N}(0,\Delta)$ and $z_t \sim \mathcal{N}(0,1)$ on the grid; all draws are independent.

Initialize $\ell^i = \ell^j = 1$, $\hat{y}_0 = \tilde{y}_0 = \mu_0$, and $t_{\text{obs}} = 0$.

for $t = 0, \Delta, \ldots, T-\Delta$ **do**

For $k \in \{i,j\}$, evaluate $m_k = m_k(\tilde{y}_t, t)$ and set $\tau_k(t) = r m_k$.

$y_{t+\Delta} \leftarrow y_t + (m_i - m_j)\Delta + \sigma \Delta W_t$ {Discretize (1)}

$\tilde{y}_{t+\Delta} \leftarrow \tilde{y}_t + (m_i - m_j)\Delta + (1 - e^{-\lambda S_t \Delta})(\hat{y}_t - \tilde{y}_t)$ {(11), using m}

$\hat{y}_{t+\Delta} \leftarrow \hat{y}_t$

for each player $k \in \{i,j\}$ and candidate $s \in \mathcal{C}_k \cap [t, t+\Delta)$, chronologically **do**

if $u_s^k < \tau_k(t)/\tau^\star$ **then**

$\hat{t}_{\ell^k}^k \leftarrow s$; $\ell^k \leftarrow \ell^k + 1$

end if

end for

if at least one submission was accepted in this step **then**

$\hat{y}_{t+\Delta} \leftarrow y_{t+\Delta} + z_t / \sqrt{\lambda(t+\Delta-t_{\text{obs}})}$; $t_{\text{obs}} \leftarrow t+\Delta$ {Readout (9)}

end if

end for

Output: (S_t), (y_t), $(\tilde{y}_t)$, $(\hat{y}_t)$, (m_i, m_j), and accepted submission times $\{\hat{t}_k^{i(j)}\}$

The first step of Algorithm 1 computes the variance path S_t and the effort policies m_i, m_j needed to simulate the contest. We obtain S_t from equation (4) and use this path to solve the nonlinear HJB system numerically, with state spacing 0.2 and time step 1/48 day. To check sensitivity to the grid, we halve both spacings; the maximum difference between the resulting effort policies is below 0.43% of peak effort on a common evaluation grid. We then simulate the contest on an hourly grid, $\Delta = 1/24$ day, using thinning (Lewis and Shedler 1979) with $\tau^\star$ bounding both submission

intensities. Accepted submissions retain their event times, whereas the public gap is refreshed only at the end of hours containing submissions and carried forward otherwise. Readout-noise variance depends on the elapsed time between public updates, not individual submissions.

Our baseline simulation is a 90-day contest with $c_i = 1.2$, $c_j = 1.5$, $\theta = 1$, $\sigma = 2$, $\lambda = 1$, $\mu_0 = 0$, $r = 15$, and $S_0 = 4$, with time measured in days. The initial variance is twice its steady-state value, $\sigma/\sqrt{\lambda} = 2$, and the incentive indices are $w_i \approx 0.208$ and $w_j \approx 0.167$. The baseline sample contains 28 submissions by player i and 18 by player j. Figure 4 displays this sample, which we also use for the estimation comparison below. A paired steady-state simulation with $S_0 = 2$, holding all other parameters, random-number streams, and simulation rules unchanged, produces 30 and 19 submissions, respectively.

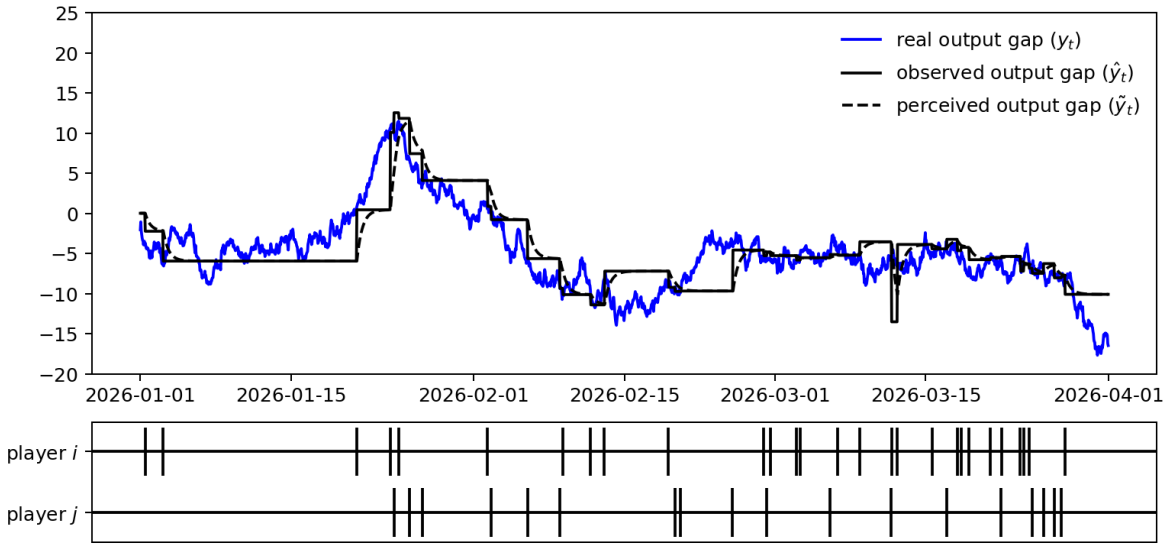


Figure 4 Synthetic Contest Data over a Three-Month Horizon

Note: The upper panel shows the latent gap, the carried-forward public gap, and the perceived gap. The lower panel records submission times for both players. Paths use the numerically computed equilibrium policy m and $S_0 = 4$.

Table 1 compares two initializations using the same complete baseline sample and jointly estimating r. Panel A imposes $S_0 = \sigma/\sqrt{\lambda}$, so the filtering variance is at its steady state from the outset. Panel B estimates S_0 separately as a nuisance parameter; its posterior is integrated out of the reported parameter summaries. The estimates are similar across initializations, and all reported 95% credible intervals contain the generating values. These results concern a single sample and do not establish the absence of systematic bias.

We compare the models primarily by their marginal likelihoods, computed by bridge sampling (Gronau et al. 2020) under common priors for the shared parameters. The Bayes factor for the non-steady-state model relative to the steady-state model is approximately one. Changing the median

Table 1 Bayesian Estimates under Alternative Initializations

Panel A. Steady-State Initialization: $S_0 = \sigma/\sqrt{\lambda}$

Parameters	c_i	c_j	σ	λ	μ_0	r
(true val.)	1.2	1.5	2.0	1.0	0.0	15.0
Posterior Mean	1.168	1.763	2.170	1.562	-0.466	13.878
Posterior SD	0.399	0.631	0.249	0.596	0.900	4.159
Post. RMSD	0.400	0.684	0.302	0.820	1.014	4.307
95% Interval	[0.58, 2.14]	[0.83, 3.34]	[1.74, 2.71]	[0.69, 2.99]	[-2.24, 1.25]	[7.53, 23.61]

Panel B. Non-Steady-State Initialization: S_0 Estimated

Parameters	c_i	c_j	σ	λ	μ_0	r
(true val.)	1.2	1.5	2.0	1.0	0.0	15.0
Posterior Mean	1.149	1.727	2.169	1.560	-0.427	13.677
Posterior SD	0.399	0.622	0.253	0.599	0.940	4.101
Post. RMSD	0.402	0.662	0.304	0.820	1.032	4.309
95% Interval	[0.57, 2.13]	[0.82, 3.21]	[1.74, 2.72]	[0.67, 3.02]	[-2.22, 1.42]	[7.41, 23.46]

Panel C. Model Comparison

	Steady State	Non-Steady State
Estimated parameters	6	7
Log marginal likelihood	-359.124	-359.127
Maximized log likelihood	-352.922	-352.717
BIC ($n = 45$)	728.684	732.080

Note: Both models use the same 46 submissions and 45 hourly public readouts, with no initial observations discarded. Both use $c_i, c_j \sim \text{LogNormal}(\log 1, 0.75^2)$, $\sigma \sim \text{LogNormal}(\log 2, 0.5^2)$, $\lambda \sim \text{LogNormal}(\log 1, 0.75^2)$, $\mu_0 \sim \mathcal{N}(0,1)$, and $\log r \sim \mathcal{N}(\log 15, 0.35^2)$. Panel B additionally uses $S_0 \sim \text{LogNormal}(\log 2, 0.75^2)$. Priors are normalized on the retained synthetic supports: $c_i, c_j \in (0.1, 5)$, $\sigma \in (0.5, 10)$, $\lambda \in (10^{-6}, 100)$, $\mu_0 \in (-20, 20)$, and $S_0 > 0.01$ when free. Each fit uses four chains with 1,000 warmup and 2,000 retained draws per chain; neither has divergences, maximum $\widehat{R} < 1.002$, and minimum bulk and tail effective sample sizes exceed 4,100 and 3,800. Post. RMSD is the square root of the posterior mean squared distance from the generating parameter, not an RMSE across datasets. Intervals are equal-tailed 95% credible intervals. BIC is $-2\hat{\ell} + k \log n$, using likelihood-only optimization on the same supports and the 45-readout convention for n, not the computational grid size. The likelihood optima approach $c_j = 5$ in both models and $S_0 = 0.01$ in Panel B; BIC is therefore reported only as a heuristic comparison.

of the lognormal prior for S_0 from 2 to 1 or 4, holding its log-scale standard deviation at 0.75, gives Bayes factors between 0.97 and 1.01. Thus, this sample provides little evidence for estimating a separate initial variance, even though it was generated with $S_0 = 4$. The auxiliary BIC favors the steady-state restriction, but boundary likelihood optima make its usual asymptotic interpretation unreliable.

4.1. Parameter Recovery Under Richer Data

Given the limited evidence for estimating a separate S_0 in Table 1, we use the more parsimonious steady-state specification to assess recovery from pooled contests and longer horizons. Data are

generated using the nonlinear equilibrium policy m and $S_0 = 4$, while estimation imposes $S_0 = \sigma/\sqrt{\lambda}$ and jointly estimates $(c_i, c_j, \sigma, \lambda, \mu_0, r)$. No initial observations are discarded.

Pooled Contests. We form nested pools of 10 and 20 independent contests under common generating parameters. Each contest draws its own initial gap, innovations, signal noise, and submission events. Estimation initializes a separate latent state for each contest and combines their likelihoods under one shared parameter vector.

In Table 2, expanding the pool from 10 to 20 contests reduces the posterior standard deviations of σ, λ, and μ_0. Uncertainty about c_i, c_j, and r changes little. Strong positive posterior correlations between r and each cost parameter indicate that these samples provide limited separation of effort costs from the submission scale.

Table 2 Bayesian Estimates from Pooled Synthetic Data

Parameters	c_i	c_j	σ	λ	μ_0	r
(true val.)	1.2	1.5	2.0	1.0	0.0	15.0
Pool of 10 Contests						
Posterior Mean	1.168	1.326	1.994	0.889	-0.195	14.477
Posterior SD	0.355	0.405	0.074	0.118	0.666	4.314
Post. RMSD	0.357	0.441	0.075	0.162	0.694	4.346
Pool of 20 Contests						
Posterior Mean	1.174	1.325	2.031	1.039	0.090	14.492
Posterior SD	0.354	0.402	0.054	0.108	0.542	4.334
Post. RMSD	0.355	0.438	0.062	0.115	0.549	4.364

Note: The nested pools contain 426 submissions (227 by i, 199 by j) and 848 submissions (450 by i, 398 by j), with 418 and 836 distinct hourly readouts. The first 10 contests form the smaller pool. The generating scale is $r = 15$, but r is estimated jointly in both pools. Priors and supports match the steady-state specification in Table 1. Each fit uses four chains with 1,000 warmup and 2,000 retained draws per chain; all $\widehat{R} < 1.002$, bulk and tail effective sample sizes exceed 2,600, and no divergences or tree-depth-limit hits occur. Post. RMSD is defined as in Table 1.

Longer Contests. We next use single contests with 180- and 360-day horizons, holding $(\theta, c_i, c_j, \sigma, \lambda, \mu_0, S_0)$ at their baseline values. We adjust the submission scale r to target comparable submission frequencies across horizons. Using equation (8), we calibrate r to match the time-averaged total submission intensity of a 90-day pilot contest with $r = 15$. Calibration uses leading-order effort policies and steady-state initialization.[9] The resulting scales, $r_{180} = 18.6484375$ and $r_{360} = 47.4375$, are used in the simulations with m and $S_0 = 4$; the resulting submission frequencies need not be identical.

In Table 3, the 360-day realization yields smaller posterior standard deviations for σ and λ than the 180-day realization. However, the posterior standard deviation of μ_0 remains close to its prior

[9] On the hourly grid, calibration uses $\frac{r_T \Delta}{T} \sum_{n=0}^{T/\Delta - 1} \sum_{k \in \{i,j\}} m_k^{(1)}(\tilde{y}_{n\Delta}, \sigma/\sqrt{\lambda}, n\Delta) \approx 0.6184661122$, with $\Delta = 1/24$ day. The target is computed from a 90-day pilot with $r = 15$, and each pilot perceived-state path uses the candidate r_T. Pilot seeds are 1000 for 90 days and $5000 + T$ for longer horizons, with offsets $0, 1, 2, 3$ for Gaussian shocks, Poisson candidates, uniform marks, and the initial gap, respectively. The absolute tolerance is 5×10^{-5} submissions per day.

value of one, whereas pooling independent contests reduces this uncertainty more substantially. These comparisons reflect the realized samples and their different submission scales, not an isolated effect of contest duration.

Table 3 Bayesian Estimates from Longer-Horizon Synthetic Data

Parameters	c_i	c_j	σ	λ	μ_0	r
Contest of 6 Months						
(true val.)	1.2	1.5	2.0	1.0	0.0	18.648
Posterior Mean	1.165	1.249	2.190	1.444	-0.362	18.230
Posterior SD	0.374	0.404	0.177	0.367	0.959	5.450
Post. RMSD	0.376	0.476	0.260	0.576	1.025	5.466
Contest of 12 Months						
(true val.)	1.2	1.5	2.0	1.0	0.0	47.438
Posterior Mean	1.074	1.445	2.085	1.033	-0.596	45.969
Posterior SD	0.327	0.442	0.096	0.189	0.921	13.654
Post. RMSD	0.350	0.446	0.128	0.191	1.097	13.732

Note: The 180-day contest contains 114 submissions (59 by i, 55 by j) and 109 distinct hourly readouts; the 360-day contest contains 235 submissions (135 by i, 100 by j) and 231 readouts. Both specifications estimate r, with $\log r \sim \mathcal{N}(\log r_T, 0.35^2)$ centered at the respective generating scale. Other priors and supports match Table 1. Each fit uses four chains with 1,000 warmup and 2,000 retained draws per chain; all $\widehat{R} < 1.002$, bulk and tail effective sample sizes exceed 3,100, and no divergences or tree-depth-limit hits occur. Post. RMSD is defined as in Table 1.

5. Empirical Application

In this section, we apply the estimation framework to real Kaggle contest data. We estimate innovation risk and public-feedback precision from submission behavior and public-leaderboard histories, and assess their empirical interpretations using private-leaderboard diagnostics. We then evaluate how feedback precision should be adjusted at the observed prize and duration, and quantify the associated relative output gains.

Following the parsimonious specification examined in Section 4, we impose the steady-state initialization $S_0 = \sigma/\sqrt{\lambda}$ and jointly estimate $(c_i, c_j, \sigma, \lambda, \mu_0, r)$ for each contest.[10] Estimation uses the leading-order policy in equation (14), so precision affects both observation noise and effort incentives through $Q_t(\lambda)$. Private-leaderboard paths, reconstructed from the private scores of historical submissions after the contest ends, are excluded from the likelihood and used to assess the estimates of λ and σ. Because final private ranks enter focal-pair selection, we interpret the evidence as diagnostic validation conditional on the selected pair, rather than as validation on a fully independent sample. Figure 5 summarizes the data-model structure and validation design.

[10] The empirical priors are $c_i, c_j \sim \text{LogNormal}(\log 1, 0.75^2)$, $\sigma \sim \text{LogNormal}(\log 2, 0.5^2)$, $\lambda \sim \text{LogNormal}(\log 1, 0.75^2)$, $r \sim \text{LogNormal}(\log 10, 0.5^2)$, and $\mu_0 \sim \mathcal{N}(\hat{y}_{t_1}, 1)$. Free positive parameters have a numerical lower bound of 0.01 and no upper bound. Centering the prior for μ_0 on the first observed public gap is an empirical-Bayes initialization, on which the posterior is conditional.

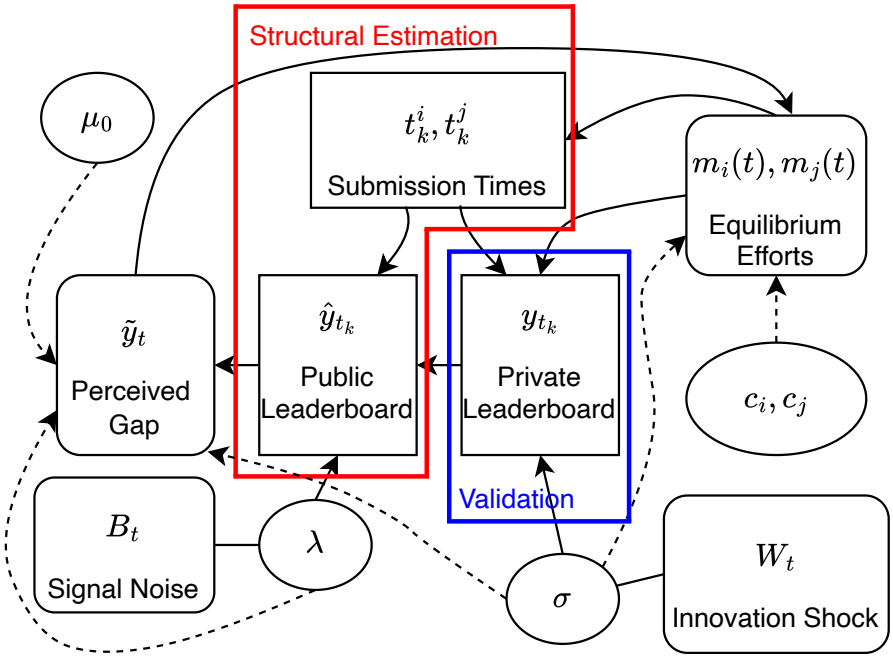


Figure 5 Meta-Kaggle Dataset and the Model Structure

Note: This figure shows the structure of a typical Kaggle contest and the information flow available to contestants. Rectangles represent observed data, rounded rectangles represent latent states, and ellipses represent contest parameters. Solid lines depict temporal relationships among time series variables and state variables, while dashed arrows show the influence of contest parameters on the data-generating process. The private gap y_t is unavailable to contestants during the contest and is used for post-estimation diagnostics.

5.1. Preparing Data for Structural Estimation

Kaggle contests differ in prize structures, participation patterns, and scoring metrics. Our contest selection follows three criteria. The resulting sample contains 73 contests concluded before April 15, 2025 that satisfy the eligibility filters on leaderboard availability, public-private score information, and focal-player activity.[11]

We first restrict attention to Kaggle competitions offering one, two, or three monetary prizes denominated in U.S. dollars. Competitions with a single prize naturally conform to the winner-take-all incentive structure. For competitions with two or three prize tiers, we use the difference between the first- and second-place cash prizes as the effective prize θ in the two-player approximation. This proxies the incentive for top rivalry, rather than the exact payoff difference for the selected focal pair.[12]

The second criterion requires an active focal pair with substantially overlapping participation. We select candidate focal contestants beginning from the top of the reconstructed final private

[11] The Meta-Kaggle database is updated daily, with new competition data added after the contests conclude.

[12] Noncash awards and separately conditioned supplementary awards, such as travel reimbursements and software-specific prizes, are excluded. Contest-specific prize conventions are reported in Appendix C.

standings. In the source records, each contestant must submit at least five times over an activity span of at least ten days, and the pair's activity windows must overlap for more than ten days. We interpret the two-player model as an approximation of rivalry among active, high-performing contestants, rather than a representation of the full participant population.

The third criterion restricts the sample to bounded, higher-is-better scoring metrics, including accuracy, AUC, average precision, Kappa, and correlation measures. To place these scores on a common transformed scale, we first map scores with range $[-1,1]$ to $[0,1]$ by $(s+1)/2$ and then apply a logit transformation:[13]

$$x_t = a \cdot \log\left(\frac{p_t}{1-p_t}\right), \quad 0 < p_t < 1 \tag{15}$$

where p_t is the clipped individual leaderboard score, x_t is the corresponding transformed score, and the scale parameter is fixed at $a = 1$ across contests. Within the clipping bounds, the transformation preserves score ordering and magnifies score differences near the endpoints of the unit interval. For the focal pair (i,j), we then construct the model variables as transformed-score gaps: $\hat{y}_t = \hat{x}_t^i - \hat{x}_t^j$ for the public leaderboard and $y_t = x_t^i - x_t^j$ for the private leaderboard.

Submission records provide calendar dates rather than within-day times. Preprocessing retains all submissions within the observation window and assigns reproducible random times within each date before constructing hourly submission counts and carried-forward score paths. These times are imputed, not observed; the estimates condition on this reconstruction. Appendix C reports implementation details and time-imputation sensitivity results.

The empirical horizon T is the length of the constructed hourly data window, including the initialization window, rather than the exact interval between the official contest opening and deadline. On this grid, we record each contestant's submission count and carry forward their latest transformed score between submissions.[14] For each hour containing at least one submission by either focal contestant, we retain one public-gap readout at the interval endpoint, following Section 3. Because submissions are asynchronous, only the submitting contestant's score is updated, while the opponent's most recent score is carried forward. The observed public gap may therefore compare scores submitted at different times and only approximates the contemporaneous latent performance gap. Our observation equation treats this gap as an approximate readout of the latent competitive state; this approximation is most credible for focal pairs with frequent submissions and substantial overlap in active periods.

[13] Preprocessing clips the rescaled scores to $[10^{-4}, 0.9999]$, affecting 54 of 26,830 public/private score entries (about 0.2%).

[14] The earliest joint public gap is backfilled into the initialization hours, which supply recursion inputs but contribute neither submission-count exposure nor validation outcomes.

5.2. Private-Leaderboard Primitive Validation

This subsection assesses whether the estimated feedback precision λ and innovation risk σ are consistent with private-leaderboard outcomes. As background, Table 4 summarizes public–private rank agreement across the 73 contests without using structural estimates. Although overall rank correlations are high, overlap among top-ranked teams is incomplete, indicating that public standings do not fully reveal private rankings.

Table 4 Public–Private Leaderboard Agreement

Statistic	Mean	Median	95% CI for Mean
Public–private rank correlation	0.937	0.984	[0.902, 0.963]
Top-2 overlap rate	65.1%	50.0%	[56.8%, 73.3%]
Top-5 overlap rate	66.6%	80.0%	[59.7%, 73.2%]

Note: The sample consists of the same 73 Kaggle contests used in the structural analysis. For each contest, we use the official public and private leaderboard ranks in Meta-Kaggle, excluding benchmark teams and teams missing either rank. Public–private rank correlation is the Spearman correlation across all retained teams within a contest. The top-k overlap rate is $|\mathcal{T}_k^{pub} \cap \mathcal{T}_k^{pri}|/k$, where $\mathcal{T}_k^{pub}$ and $\mathcal{T}_k^{pri}$ are the sets of teams ranked in the public and private top k, respectively; the ordering of teams within the top-k sets need not coincide. Each statistic is first calculated separately by contest, and the table reports the equally weighted mean and median across the 73 contests. The 95% confidence intervals pertain to the across-contest means and are percentile intervals obtained from 10,000 nonparametric bootstrap resamples at the contest level.

For each of the 73 contests, we estimate the model using the full observation window and again using only the first 75% of the contest-phase window for the time-split diagnostics, retaining the full-sample horizon in both estimations. The 73 full-sample fits and 73 early-sample fits all have no divergent transitions and $\widehat{R} < 1.01$. Appendix C reports the full-sample parameter estimates.

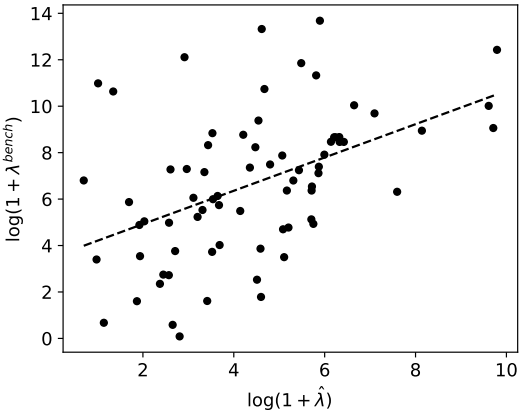
We next examine whether contests with higher estimated λ exhibit less public–private disagreement and those with higher estimated σ exhibit greater private-gap volatility. Appendix C provides the regression details.

5.2.1. Private-Leaderboard Diagnostic for Signal Precision λ. We assess the estimated feedback precision $\hat{\lambda}_\ell$ by comparing it with a benchmark constructed from public–private leaderboard discrepancies. Let $\ell \in \{1, \ldots, L\}$ index contests. Let $\mathcal{E}_\ell$ denote the ordered set of submission-triggered leaderboard readouts for the focal pair, and let $N^\ell = |\mathcal{E}_\ell|$. Let $h_k^\ell = t_k^\ell - t_{k-1}^\ell$ denote the elapsed time associated with readout t_k^ℓ. Using the Gaussian discrepancy approximation $\hat{y}_{t_k}^\ell - y_{t_k}^\ell \sim \mathcal{N}(0, 1/(\lambda_\ell h_k^\ell))$ for $t_k^\ell \in \mathcal{E}_\ell$, we construct

$$\lambda_\ell^{\text{bench}} = \frac{N^\ell}{\sum_{t_k^\ell \in \mathcal{E}_\ell} h_k^\ell (y_{t_k}^\ell - \hat{y}_{t_k}^\ell)^2}.$$

The benchmark therefore uses only submission-triggered leaderboard readouts rather than repeated hourly carry-forward observations.[15] This working Gaussian law does not follow from the score transformation, and asynchronous updates may induce dependence across readouts. The benchmark therefore checks the observation-precision interpretation of λ, not its behavioral effect through $Q_t(\lambda)$ and effort.

Figure 6 shows a positive association between the two precision measures (slope 0.717; Spearman $\rho = 0.508$, $p < 0.001$). Because the two measures share public-data inputs, their positive association is a directional consistency check rather than an independent holdout validation.



	coef.	stderr	95% CI
const.	3.490***	0.868	[1.790, 5.191]
$\log(1+\hat{\lambda})$	0.717***	0.156	[0.412, 1.022]
Obs.		73	
R^2		0.218	
Adj. R^2		0.207	

(1) Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. (2) OLS residual diagnostics: Omnibus = 3.961, Jarque–Bera = 3.281, Skewness = 0.506, Kurtosis = 3.235. (3) Condition Number = 12.122.

Figure 6 $\lambda_\ell^{\text{bench}}$ vs. Bayesian Estimate $\hat{\lambda}_\ell$

Note. Each point represents one of 73 contests. The approximate benchmark $\lambda_\ell^{\text{bench}}$ uses elapsed-time-weighted squared public–private discrepancies at submission-triggered readouts; $\hat{\lambda}_\ell$ is the posterior mean structural estimate. The dashed line regresses $\log(1+\lambda_\ell^{\text{bench}})$ on $\log(1+\hat{\lambda}_\ell)$.

5.2.2. Public-Private Leaderboard Disagreement Diagnostics. Whereas the preceding diagnostic compares two precision measures, we now examine whether higher estimated precision is associated with less public–private disagreement. For contest ℓ, we measure disagreement by the mean absolute gap between the two leaderboard paths:

$$D_\ell^{path} = \frac{1}{|\mathcal{H}_\ell|} \sum_{h \in \mathcal{H}_\ell} |y_{\ell h} - \hat{y}_{\ell h}|,$$

where $\mathcal{H}_\ell$ contains the hourly observation points in the contest-phase path, excluding the backfilled initialization window. Higher λ reduces signal noise at a given readout interval, motivating a negative association with D_ℓ^{path}; update frequency and score characteristics also affect observed disagreement.

[15] As a robustness comparison to the irregular-interval estimator above, we also consider the fixed-grid approximation $\lambda_{\ell,\Delta}^{\text{bench}} = N^\ell / \left\{ \Delta \sum_{t_k^\ell \in \mathcal{E}_\ell} (y_{t_k}^\ell - \hat{y}_{t_k}^\ell)^2 \right\}$. Despite this difference, both constructions yield the same qualitative conclusion of a positive but modest cross-contest association.

Because $\hat{\lambda}_\ell$ and D_ℓ^{path} use public data from the same period, their association may partly reflect shared path variation. We therefore also relate the early estimate $\hat{\lambda}_\ell^{early}$ to disagreement in the final quarter. Let $\mathcal{H}_\ell^{late}$ contain the hours in the final 25% of the contest-phase window and define $D_\ell^{late} = |\mathcal{H}_\ell^{late}|^{-1} \sum_{h \in \mathcal{H}_\ell^{late}} |y_{\ell h} - \hat{y}_{\ell h}|$. Initialization hours are excluded when locating the cutoff.

We regress $\log(1 + D_\ell^{path})$ on $\log(1 + \hat{\lambda}_\ell)$ for the full-path comparison and $\log(1 + D_\ell^{late})$ on $\log(1 + \hat{\lambda}_\ell^{early})$ for the time-split comparison. Table 5 reports negative slopes for both the full path (-0.022, $p = 0.0012$) and the time split (-0.024, $p = 0.019$). Thus, higher estimated precision is associated with less disagreement not only over the same period, but also when early estimates are compared with final-quarter outcomes. The time-split relationship has limited explanatory power ($R^2 = 0.038$).

Table 5 Signal Precision and Public-Private Path Disagreement

Specification	coef.	stderr	95% CI	R^2	Spearman ρ
Full path	-0.022**	0.007	[-0.036, -0.009]	0.101	-0.423***
Time split	-0.024*	0.010	[-0.044, -0.004]	0.038	-0.277*
Obs.			73		

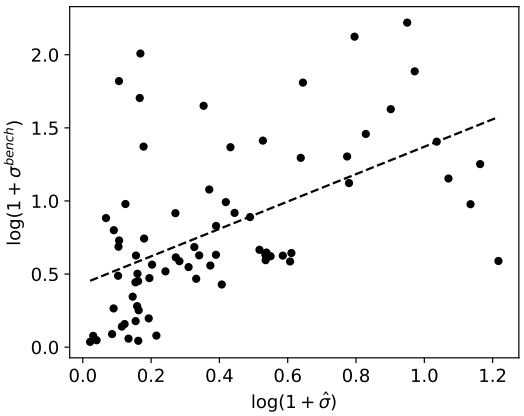
Note: The full-path row regresses $\log(1 + D_\ell^{path})$ on $\log(1 + \hat{\lambda}_\ell)$, where D_ℓ^{path} is mean absolute public–private gap disagreement. The time-split row relates the same measure in the final 25% to precision estimated from the first 75% of contest-phase data, retaining the full-sample horizon. Initialization hours are excluded from the discrepancy measures. Both regressions include an intercept and use 73 contests and posterior mean estimates. Spearman ρ reports the corresponding rank association. (1) Significance markers: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. (2) Residual-distribution diagnostics: Omnibus and Jarque–Bera test residual normality; skewness and kurtosis describe asymmetry and tail heaviness. Values are reported as Full path; Time split: Omnibus = 63.469; 87.472, Jarque–Bera = 319.558; 1001.859, Skewness = 2.719; 3.686, Kurtosis = 11.689; 19.584. (3) Condition Number (Full path; Time split): 12.122; 10.572.

5.2.3. Private-Leaderboard Consistency Benchmark for Innovation Risk σ. We assess the estimated innovation risk $\hat{\sigma}_\ell$ using private-gap increments adjusted for estimated effort drift. Let $(t_k^\ell, y_k^\ell)_{k=0}^{n_\ell}$ denote the private-leaderboard gap evaluated at the ordered update times used for this benchmark, with the first retained readout serving as the initial observation. No increment is constructed from the artificial backfill. Motivated by a Gaussian increment approximation to equation (1), we subtract the posterior-estimated effort drift $\hat{\mu}_k^\ell = \int_{t_{k-1}^\ell}^{t_k^\ell} [\hat{m}_i^\ell(s) - \hat{m}_j^\ell(s)]\, ds$ from each increment and scale the squared residual by the elapsed time. The resulting benchmark is

$$\sigma_\ell^{\text{bench}} = \left(\frac{1}{n_\ell} \sum_{k=1}^{n_\ell} \frac{\left[y_k^\ell - y_{k-1}^\ell - \hat{\mu}_k^\ell \right]^2}{t_k^\ell - t_{k-1}^\ell} \right)^{1/2}$$

Appendix B.4 derives this expression. The drift adjustment $\hat{\mu}_k^\ell$ is evaluated at the posterior mean parameters. Together with asynchronous score updates, this makes $\sigma_\ell^{\text{bench}}$ a model-assisted consistency benchmark, rather than an exact maximum likelihood estimator of latent diffusion volatility.

As for λ, we examine the cross-contest relationship between $\log(1+\sigma_\ell^{\text{bench}})$ and $\log(1+\hat{\sigma}_\ell)$. The positive association in Figure 7 (slope 0.937; Spearman $\rho = 0.566$, $p < 0.001$) is consistent with the innovation-risk interpretation of $\hat{\sigma}_\ell$.



	coef.	stderr	95% CI
const.	0.434***	0.096	[0.246, 0.622]
$\log(1+\hat{\sigma})$	0.937***	0.209	[0.526, 1.347]
Obs.		73	
R^2		0.282	
Adj. R^2		0.271	

(1) Significance levels: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. (2) OLS residual diagnostics: Omnibus = 14.614, Jarque–Bera = 16.118, Skewness = 1.054, Kurtosis = 3.925. (3) Condition Number = 3.781.

Figure 7 σ^{bench} **vs. Bayesian Estimate** $\hat{\sigma}$

Note. Each point represents one of 73 contests. The model-assisted benchmark $\sigma_\ell^{\text{bench}}$ uses private-gap increments between readouts, adjusted for elapsed time and effort drift evaluated at posterior mean parameters; $\hat{\sigma}_\ell$ is the posterior mean structural estimate. The dashed line regresses $\log(1+\sigma_\ell^{\text{bench}})$ on $\log(1+\hat{\sigma}_\ell)$.

5.2.4. Private-Leaderboard Volatility Diagnostics. We examine whether contests with higher estimated innovation risk exhibit greater private-path volatility. Unlike the preceding benchmark, this diagnostic does not use an estimated effort-drift adjustment. For contest ℓ, let $\Delta y_{\ell h} = y_{\ell h} - y_{\ell,h-1}$ denote the hourly private-gap increment and define $V_\ell^{path} = \text{sd}\{\Delta y_{\ell h} : h \in \mathcal{H}_\ell,\ h > 1\}$. Hours are indexed within the contest-phase path, excluding artificial initialization increments.

Carrying scores forward produces zero increments between updates and jumps when new scores arrive. Thus, V_ℓ^{path} reflects both submitted-score changes and update frequency. It provides a less model-dependent check on $\hat{\sigma}_\ell$, rather than a direct measure of latent diffusion volatility.

We also apply the same time split used for signal precision. For the final-quarter hours $\mathcal{H}_\ell^{late}$, define $V_\ell^{late} = \text{sd}\{\Delta y_{\ell h} : h \in \mathcal{H}_\ell^{late}\}$. The full-path regression relates $\log(1+V_\ell^{path})$ to $\log(1+\hat{\sigma}_\ell)$; the time-split regression relates $\log(1+V_\ell^{late})$ to $\log(1+\hat{\sigma}_\ell^{early})$.

Table 6 reports positive slopes for both the full path (0.238, $p < 0.001$) and the time split (0.301, $p = 0.004$). The association with innovation risk therefore also appears without the benchmark's drift adjustment and when early estimates are compared with later volatility.

The preceding diagnostics support the empirical interpretations of λ and σ, but do not establish the accuracy of the leading-order effort approximation. The incentive indices $w_k = \theta/(\sigma^2 c_k)$ are not uniformly small across the sample, so Theorem 1 alone cannot establish this accuracy. We therefore interpret the following counterfactuals as implications of the leading-order specification, rather than predictions from the exact nonlinear equilibrium.

Table 6 Private-Leaderboard Volatility Diagnostics

Specification	coef.	stderr	95% CI	R^2	Spearman ρ
Full path	0.238***	0.057	[0.126, 0.349]	0.247	0.641***
Time split	0.301**	0.106	[0.094, 0.508]	0.256	0.368**
Obs.			73		

Note: The full-path row regresses $\log(1+V_\ell^{path})$ on $\log(1+\hat{\sigma}_\ell)$, where V_ℓ^{path} is the sample standard deviation of hourly private-gap increments within the contest-phase path. The time-split row uses increments ending in the final 25% and innovation risk estimated from the first 75%, retaining the full-sample horizon. Artificial initialization increments are excluded. Both regressions include an intercept and use 73 contests and posterior mean estimates. Spearman ρ reports the corresponding rank association. (1) Significance levels: *** $p<0.001$, ** $p<0.01$, * $p<0.05$. (2) OLS residual diagnostics (Full path; Time split): Omnibus = 43.382; 22.369, Jarque–Bera = 116.544; 37.414, Skewness = 1.990; 1.141, Kurtosis = 7.741; 5.664. (3) Condition Number (Full path; Time split): 3.781; 3.948.

5.3. Implication for Precision Design

We now examine feedback precision design in the Kaggle contests using the estimated primitives. For each contest, we assess whether feedback should become more or less precise and how much output could improve.

We retain the observed prize and duration, so prize expenditure and the delay cost $D(T)$ remain unchanged. With no precision costs, we use posterior mean output as the design criterion, without specifying the functional forms of u or D. For each contest ℓ, the recommended precision is

$$\lambda_\ell^* \in \arg\max_{\lambda\in(0,\infty]} \mathbb{E}\left[\mathcal{B}_\ell^{\mathrm{cal}}\left(\theta_\ell^{obs}, T_\ell^{obs}, \lambda\right)\middle|\, \mathrm{data}_\ell\right],$$

where $\mathcal{B}_\ell^{\mathrm{cal}}$ denotes expected output under the steady-state calibration and $\lambda=\infty$ represents full disclosure. This exercise compares steady-state feedback regimes, rather than the fixed-S_0 joint designs in Proposition 1. Appendix C.1 reports the calibration and calculation details.

Table 7 distinguishes support for an adjustment from its expected benefit. Panel A reports whether increasing or reducing precision has more than 75% posterior support. Panel B summarizes relative output gains from adopting the single recommended precision λ_ℓ^* in each contest.

The recommended precision is full disclosure in 67 contests and finite in six. Of the six, three have more than 75% posterior support for reducing precision; the remaining three have no clear adjustment direction under this threshold. The median relative output gain is 0.156% across all contests and 1.776% among the three supporting reduced precision.

6. Robustness

This section addresses the focal-pair selection and Gaussian-approximation concerns raised in Section 3.2. We examine alternative rivalries and, as an indirect check within the maintained Gaussian framework, alternative score calibrations.

Table 7 Precision Adjustments and Expected Output Gains

Panel A: Posterior support for precision adjustment

Adjustment direction	Contests	Share (%)
Increase precision	67	91.8
Decrease precision	3	4.1
Direction uncertain	3	4.1

Panel B: Expected output gains (%)

Contest group	25th percentile	Median	75th percentile
All contests	0.038	0.156	0.441
Increase precision	0.042	0.147	0.399
Decrease precision	1.009	1.776	2.967
Direction uncertain	0.014	0.028	1.730

Note: The sample contains 73 contests. Panel A compares draw-specific optimal precision with current precision from the same posterior draw; a direction is reported when its probability exceeds 75%. Panel B reports cross-contest percentiles of the percentage change in posterior mean output from current precision to the recommended precision λ_ℓ^*. Prize and duration remain fixed. Percentiles describe variation across contests, not credible intervals.

6.1. Selection of Two Contestants

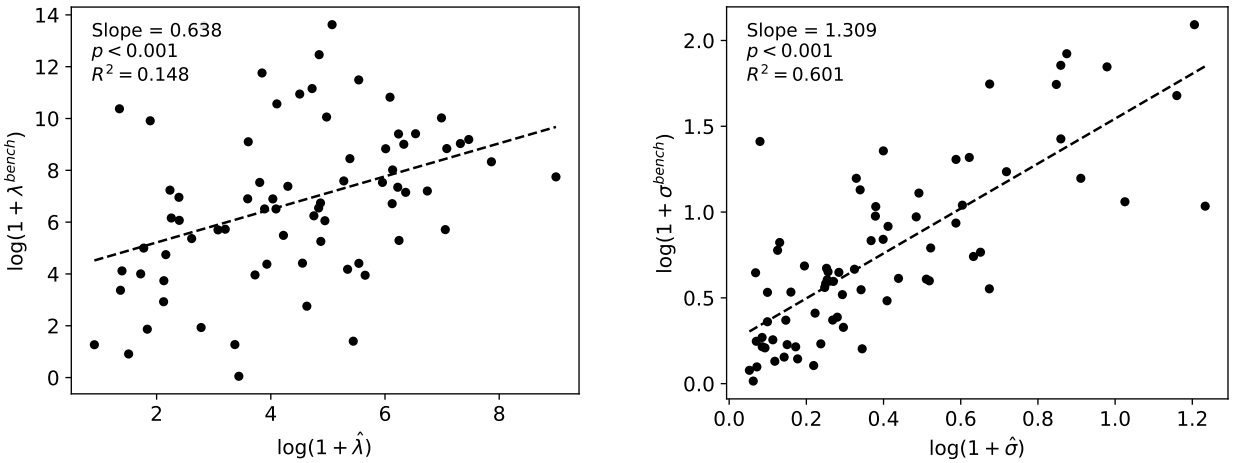
We exclude the baseline contestant i to assess whether the validation patterns depend on this contestant. Contestants are ranked by their last recorded private score, and we search the remaining candidates within the top 20 for an alternative pair satisfying the activity and overlap requirements in Section 5.1. All 73 contests admit an alternative pair, with the baseline prize, horizon, and preprocessing rules retained.

Figure 8 repeats the private-leaderboard benchmark comparisons using these pairs. The associations for both λ and σ remain positive and statistically significant, preserving the qualitative validation patterns under alternative focal-pair selection. Appendix C.2 reports the detailed statistics and the resulting changes in effective observation windows.

6.2. Normal Approximation

The Gaussian assumption in Section 3.2 makes posterior beliefs tractable through their mean and variance and supports the effort policy, likelihood, and validation benchmarks. Replacing it generally changes belief updating and hence contestants' effort choices. A meaningful non-Gaussian assessment would therefore require rederiving these components and re-estimating the model jointly, an extension we do not undertake here.

As an indirect check, we calibrate the score-scale parameter a in equation (15) to a standard normal reference under L_1 and L_2 losses and re-estimate all 73 contests, retaining the baseline prior specifications in transformed units. Although adjustment probabilities and output gains vary, the rankings of λ and σ have correlations above 0.99 with the baseline, and every contest retains its full-disclosure versus finite-precision classification (67 and six contests, respectively). This checks



(a) Signal precision (b) Innovation risk

Figure 8 Private-Leaderboard Validation with Alternative Focal Pairs

Note: Each panel contains 73 contests. Benchmarks are constructed as in Section 5.2, using the actual intervals between retained readouts. Dashed lines are fitted regressions of the log-transformed benchmarks on the corresponding log-transformed posterior mean estimates. Reported p-values use HC1 heteroskedasticity-robust standard errors.

sensitivity to score calibration, not Gaussianity itself: changing a does not alter standardized distributional shape. Appendix C.2 provides the details.

7. Conclusion

This paper develops a dynamic game-theoretic model and Bayesian structural estimation framework for online crowdsourcing contests under noisy public feedback, with the final private leaderboard determining the realized winner. Contestants evaluate the prize using their full posterior winning probabilities, yielding a closed-form leading-order effort policy in which signal precision determines how broadly incentives extend across public performance gaps. We prove that a finite-precision design can strictly outperform every full-disclosure design even without a direct precision cost.

Conditional on focal pairs selected using final private ranks, we recover latent competition states, effort dynamics, and contest-specific primitives from contest-phase submission timing and public leaderboard trajectories, and assess the estimated primitives using private-leaderboard path values excluded from the likelihood. Across 73 Kaggle contests, higher estimated innovation risk is associated with greater private-leaderboard volatility, while higher estimated feedback precision is associated with less public–private disagreement. Both relationships are supported by time-split tests using later private-leaderboard outcomes. At the observed prize and duration, the steady-state, leading-order calibration yields contest-specific precision recommendations. Most contests favor increased precision under this calibration, although the median relative output gain is modest.

Several limitations deserve mention. The incentive indices are not uniformly small across the empirical sample, so the leading-order effort approximation lacks a uniform accuracy guarantee.

The two-player framework simplifies multi-tier prizes and asynchronous feedback, while empirical estimation maintains Gaussian updating and fixes the filtering variance at its parameter-dependent steady state. The precision recommendations are therefore calibrated benchmarks rather than exact joint-design predictions. Proposition 1 concerns symmetric contestants and a zero initial perceived gap. Its sufficient results do not constitute a complete partition of the parameter plane; the broader regime patterns and plotted boundaries are numerical findings. These limitations suggest natural extensions involving nonlinear structural estimation, richer prize structures, nonstationary filtering, and multi-agent dynamics.

Appendix A: Proofs

Proof of Theorem 1 Fix a finite $\lambda > 0$. Put $a_t = \lambda S_t^2$, $\alpha_k = \sigma^2 w_k$, and $\alpha_* = \max_k \alpha_k$. Write $U^k = V^k/\theta$, $p = U^i_y$, and $n = -U^j_y$. In reverse time $\tau = T - t$, the differentiated HJB equations are

$$p_\tau = \tfrac{a_{T-\tau}}{2} p_{yy} + \partial_y(\alpha_i p^2/2 - \alpha_j pn), \qquad n_\tau = \tfrac{a_{T-\tau}}{2} n_{yy} + \partial_y(\alpha_i pn - \alpha_j n^2/2),$$

with $p(y,0) = n(y,0) = G_{S_T}(y)$, where G_q is the centered Gaussian density of variance q.

Existence. We apply the integral-equation version of the abstract Cauchy–Kowalevski theorem in Nishida (1978, Chapter III, Remark 2.3). Use the analytic norms

$$\|f\|_\rho = (2\pi)^{-1} \int_{\mathbb{R}} e^{\rho|\xi|} |\widehat{f}(\xi)|\, d\xi, \qquad \widehat{f}(\xi) = \int_{\mathbb{R}} e^{-i\xi y} f(y)\, dy.$$

For $0 < \rho' < \rho$, these satisfy $\|fg\|_\rho \le \|f\|_\rho \|g\|_\rho$ and $\|f_y\|_{\rho'} \le \|f\|_\rho/(\rho - \rho')$. Let $P_{t,s}$ be Gaussian convolution of variance $\int_t^s a_u\, du$. It is a contraction in every such norm. For $h = (p,n)$, let $\mathcal{N}(h)$ denote the two nonlinear terms above and use the sum norm for pairs. The gradient equations have the integral form

$$h(\tau) = P_{T-\tau,T} h(0) + \int_0^\tau P_{T-\tau,T-s} \mathcal{N}(h(s))\, ds. \tag{16}$$

On a radius-R ball,

$$\|\mathcal{N}(h) - \mathcal{N}(\widehat{h})\|_{\rho'} \le \frac{C\alpha_* R}{\rho - \rho'} \|h - \widehat{h}\|_\rho, \qquad \|\mathcal{N}(h)\|_{\rho'} \le \frac{C\alpha_* R^2}{\rho - \rho'}.$$

Fix $\rho_0 > \rho_1 > 0$ and $R > 2\|h(0)\|_{\rho_0}$; the Gaussian data make this choice independent of α_*. After rescaling time by $r = \alpha_* \tau$, the estimates above have constants independent of α_*, since $\alpha_k/\alpha_* \le 1$ and the propagators remain contractions. The cited theorem thus gives an existence interval $0 \le r < r_0$ and a bound in the ρ_1 norm, both independent of α_*. Because $\alpha_* \le \sigma^2(w_i + w_j)$, taking $w_i + w_j$ sufficiently small ensures $\alpha_* T < r_0$. The solution then exists on $[0,T]$ with a fixed positive analytic radius. Cauchy's estimate on any smaller strip gives uniform bounds on all spatial derivatives used below.

Equilibrium verification. Returning to $t = T - \tau$, with $b = \alpha_i p - \alpha_j n$, the gradients satisfy

$$p_t + \tfrac{a_t}{2} p_{yy} + b p_y - \alpha_j n_y p = 0, \qquad n_t + \tfrac{a_t}{2} n_{yy} + b n_y + \alpha_i p_y n = 0.$$

Their bounded coefficients and strictly positive Gaussian terminal data give $p, n > 0$ by Feynman–Kac. Bounded drift and reaction coefficients also preserve Gaussian tail bounds. Integrating the reverse-time conservative equations over $\mathbb{R}$ therefore produces no boundary terms and preserves $\int p\, dy = \int n\, dy = 1$. Consequently, $U^i(y,t) = \int_{-\infty}^y p(z,t)\, dz$ and $U^j(y,t) = \int_y^\infty n(z,t)\, dz$ are bounded values with the required terminal and spatial limits. Integrating the gradient equations fixes the possible constants of integration by these spatial limits and gives the normalized HJB system $U^k_t + (a_t/2) U^k_{yy} + F_k(U) = 0$, where

$$F_i(U) = \alpha_i (U^i_y)^2/2 + \alpha_j U^i_y U^j_y, \qquad F_j(U) = \alpha_j (U^j_y)^2/2 + \alpha_i U^i_y U^j_y.$$

The positive policies $m_i = \alpha_i p$ and $m_j = \alpha_j n$ attain the constrained HJB maxima. For any admissible deviation q_k, Itô's formula and completion of the square give

$$\mathbb{E}_{y,t}\left[G_k(\tilde{y}_T) - \int_t^T \frac{c_k q_{k,s}^2}{2}\, ds \right] = V^k(y,t) - \frac{c_k}{2} \mathbb{E}_{y,t} \int_t^T [q_{k,s} - m_k(\tilde{y}_s, s)]^2\, ds. \tag{17}$$

The stochastic integral has zero expectation by bounded derivatives and admissibility. The last term in (17) is nonpositive and vanishes for $q_k = m_k$. Hence the policies are mutual best responses and form an interior Markov-perfect equilibrium.

Leading-order effort. Direct differentiation, using $Q_t' = -a_t$ and $Q_T = S_T$, verifies $U_i^{(0)} = \Phi(y/\sqrt{Q_t})$, $U_j^{(0)} = 1 - U_i^{(0)}$ as the zero-incentive HJB solution. The exact values satisfy

$$U^k(t) = U_k^{(0)}(t) + \int_t^T P_{t,s} F_k(U(s))\, ds. \tag{18}$$

Indeed, $Q_t - Q_s = \int_t^s a_u\, du$, so propagating the terminal density G_{S_T} from T to t gives G_{Q_t}. On a smaller fixed analytic strip, the uniform existence bound controls $\mathcal{N}(h)$ by $C\alpha_*$. Heat contraction and integration over an interval of length at most T therefore give

$$\sup_{y,t}\{|p(y,t) - G_{Q_t}(y)| + |n(y,t) - G_{Q_t}(y)|\} \le CT\alpha_* = O(w_i + w_j).$$

Since $m_k = \alpha_k p_k = \sigma^2 w_k p_k$, multiplying by $\sigma^2 w_k$ gives the stated expansion. Finally, $Q_t' = -a_t < 0$, $Q_T = S_T$, and $Q_0 = S_0 + \sigma^2 T$, so $S_T \le Q_t \le S_0 + \sigma^2 T$; the Gaussian factor is therefore uniformly bounded. This proves (5) uniformly in state and time. □

Proof of Proposition 1 The utility assumption implies $u'(B) \le Ce^{-\zeta B}$ for all sufficiently large B, so u is bounded above and $Bu'(B) \to 0$. Write $\alpha = \theta/c$, $b_0 = \sqrt{S_0/(2\pi)}$, and $b(T) = \sqrt{(S_0 + \sigma^2 T)/(2\pi)}$; put $Q_t = S_t + \sigma^2(T - t)$.

Part (i): finite feedback strictly dominates full disclosure. Fix $\bar\lambda, \bar\alpha > 0$. The analytic existence argument in the proof of Theorem 1 is uniform for $0 < \sigma \le \bar\sigma$ and sufficiently small T. Indeed, $S_t \ge S_0/(1 + \bar\lambda S_0 t)$ keeps the Gaussian terminal data bounded on a common analytic strip. With $\alpha_i = \alpha_j = \bar\alpha$, the gradient integral equation gives $m_k(y,t) = \bar\alpha G_{Q_t}(y) + O(T)$ uniformly in state and time.

Under zero effort, $\tilde y_t$ is centered Gaussian with variance $Q_0 - Q_t$. Gaussian convolution therefore gives $\mathbb{E}[G_{Q_t}(\tilde y_t)] = G_{Q_0}(0)$. Moreover, the leading terms of m_i and m_j coincide, so $m_i - m_j = O(T)$; integrating this drift over a horizon of length T changes the perceived-gap path by $O(T^2)$. The Lipschitz Gaussian kernels give the same order of error after taking expectations. Substitution in (26) first gives

$$\mathcal{B}(0, c\bar\alpha, T, \bar\lambda; S_0) = b(T) + \bar\alpha G_{Q_0}(0) T + o(T).$$

Since $G_{Q_0}(0) = 1/\sqrt{2\pi(S_0 + \sigma^2 T)}$, $b'(0) = \sigma^2/\sqrt{8\pi S_0}$, and $D'(0) = 0$, Taylor expansion gives, uniformly in $0 < \sigma \le \bar\sigma$,

$$u(\mathcal{B}) - D(T) = u(b_0) + \frac{u'(b_0)(\bar\alpha + \sigma^2/2)}{\sqrt{2\pi S_0}} T + o(T).$$

The displayed coefficient is bounded below by $u'(b_0)\bar\alpha/\sqrt{2\pi S_0} > 0$. Uniformity of the remainder lets us choose a single short $\bar T > 0$ and $\delta > 0$, independent of $\sigma \in (0, \bar\sigma]$, such that the payoff before deducting the prize is at least $u(b_0) + \delta$. The prize equals $\bar\theta = c\bar\alpha$. Hence, for $c < c_F := \delta/(4\bar\alpha)$, this finite-precision candidate has net payoff greater than $u(b_0) + 3\delta/4$.

Use the full-disclosure benchmark in Appendix B.1, with $w = \alpha/\sigma^2$.

Couple the equilibrium scores with their zero-effort counterparts using the same Brownian paths. The maximum function is one-Lipschitz in the Euclidean norm, and $(\int_0^T m_k dt)^2 \le T\int_0^T m_k^2 dt$. Conditional Cauchy–Schwarz and (25) therefore give, for $Y_0 \sim \mathcal{N}(0, S_0)$,

$$\mathcal{B}(0,\theta,T,\infty;S_0) - b(T) \le \sqrt{2\alpha T}\,\mathbb{E}\sqrt{1-f(d(Y_0,0))}.$$

For $d>0$ and $x = wd/2$,

$$\frac{wf'(d)K(d)}{K'(d)} = 1 + (x\tanh x - 1)\operatorname{sech}^2 x \le 2,$$

using $\cosh^2 x \ge 1 + x^2 \ge 2x$. Because $f(1)=1$ and $K, K' > 0$ on $(0,1]$, the ratio bound implies

$$1 - f(d) = \int_d^1 f'(s)ds \le \frac{2}{w}\int_d^1 \frac{K'(s)}{K(s)}ds = \frac{2}{w}\log\frac{K(1)}{K(d)}.$$

Equation (24) and symmetry in y therefore yield

$$1 - f(d(y,0)) \le -\frac{2}{w}\log\left|2\Phi\left(\frac{y}{\sigma\sqrt{T}}\right) - 1\right|.$$

The constant $C := 2\int_{\mathbb{R}} \sqrt{-\log|2\Phi(z)-1|}\,dz$ is finite: its integrand is $O(\sqrt{|\log|z||})$ near zero and has Gaussian decay in the tails. Since $w = \alpha/\sigma^2$, the preceding bounds imply

$$\mathcal{B} - b(T) \le 2\sigma\sqrt{T}\,\mathbb{E}\sqrt{-\log|2\Phi(Y_0/(\sigma\sqrt{T})) - 1|}.$$

Changing variables $Y_0 = \sigma\sqrt{T}z$ and bounding its Gaussian density by $1/\sqrt{2\pi S_0}$ gives

$$\mathcal{B}(0,\theta,T,\infty;S_0) \le b(T) + \frac{C\sigma^2 T}{\sqrt{2\pi S_0}}. \tag{19}$$

This also holds at $\theta = 0$ or $T = 0$.

Concavity of the square root gives $b(T) \le b_0 + \sigma^2 T/(2\sqrt{2\pi S_0})$. Concavity of u, followed by (19) and omission of the nonpositive term $-\theta$, gives, for every full-disclosure prize and deadline,

$$u(\mathcal{B}) - \theta - D(T) \le u(b_0) + a\sigma^2 T - D(T), \qquad a = \frac{(C+1/2)u'(b_0)}{\sqrt{2\pi S_0}}.$$

Since $D'(T)/T$ is nondecreasing and converges to $D''(0)$ as $T \downarrow 0$, $D'(T) \ge D''(0)T$. Integrating from zero gives $D(T) \ge D''(0)T^2/2$, so completing the square yields

$$\sup_{T\ge 0}\{a\sigma^2 T - D(T)\} \le \frac{a^2\sigma^4}{2D''(0)}.$$

Choose $\sigma_F \le \min\{\bar\sigma, 1\}$ so that the last bound is strictly below $\delta/2$. Every full-disclosure payoff is then below $u(b_0) + \delta/2$, whereas the constructed candidate is above $u(b_0) + 3\delta/4$. Thus the candidate strictly beats every full-disclosure design for $0 < c < c_F$ and $0 < \sigma < \sigma_F$.

When $\theta = 0$ or $T = 0$, the payoff is precision-independent, so the same payoff is attained by setting $\lambda = \infty$; these boundary designs are already included in the full-disclosure comparison. Hence no joint optimizer can have zero prize, zero deadline, or full disclosure. Every joint optimizer therefore lies in $(0,\infty)^3$. This proves part (i) without asserting that a joint optimizer exists.

Part (ii): positive-prize full disclosure. Take (c, σ, T_0, b, V_0) from Lemma 3; $c < c_e$ and $\sigma > 1 \ge \sigma_F$. Let $u_\infty := \lim_{B\to\infty} u(B) < \infty$. Any full-disclosure design with payoff at least V_0 satisfies $\theta + D(T) \le u_\infty - V_0$. Since

D is strictly increasing and unbounded, such designs lie in a compact prize–deadline rectangle. The explicit policies are continuous before $T-\varepsilon$, and (25) controls the remaining score change by $O(\sqrt{\varepsilon})$, uniformly on bounded α, including zero prize and zero deadline. It follows by first taking parameter limits on $[0, T-\varepsilon]$ and then sending $\varepsilon \downarrow 0$ that the objective extends continuously to this rectangle. It therefore attains its maximum (Berge 1963, Chapter 6).

Rescaling (29) at $T = T_0$ gives

$$\partial_\theta \{u(\mathcal{B}) - \theta - D(T_0)\}|_{0+} = \frac{T_0 u'(b)}{c\sqrt{2\pi(S_0 + \sigma^2 T_0)}} - 1 = \frac{c_e}{c} - 1 > 0. \tag{20}$$

The positive right derivative means that a sufficiently small prize at T_0 yields payoff strictly above V_0; hence the attained full-disclosure maximum exceeds V_0. Lemma 3 excludes every positive-prize finite-feedback design; zero-prize and zero-deadline designs yield at most V_0 and $u(b_0)$. Consequently this is a joint maximum, and every joint maximizer satisfies $\theta^* > 0$, $T^* > 0$, and $\lambda^* = \infty$.

Part (iii): high-cost prize exit. Fix $\sigma > 0$. Let T_0 maximize $u(b(T)) - D(T)$ and write $V_0 = u(b(T_0)) - D(T_0)$, $\Gamma = \lim_{B\to\infty} u(B) - u(b_0) > 0$. Choose $\bar{T}$ from $D(\bar{T}) = \Gamma$. Every design with payoff at least V_0 has $\theta + D(T) \le \Gamma$: indeed, $u(\mathcal{B}) - \theta - D(T) \ge V_0 \ge u(b_0)$ and $u(\mathcal{B}) \le \lim_{B\to\infty} u(B)$. Hence $T \le \bar{T}$ and $\alpha \le \Gamma/c \to 0$ uniformly. We bound the output gain directly, without ranking feedback precision.

For $\lambda S_0^2 < .72\sigma^2$, $S_T \ge S_0$ and $\alpha T/S_T \le \alpha\bar{T}/S_0 \to 0$. Thus (31) applies uniformly for arbitrarily small positive precision. For $\lambda S_0^2 \ge .72\sigma^2$, use (30) with $w = \alpha/\sigma^2 \to 0$. For full disclosure, (25) and $\log\cosh z \le z^2/2$ give

$$1 - f(d) \le \frac{2}{w}\log\cosh(w/2) \le \frac{w}{4}, \qquad \mathbb{E}\int_0^T (m_i^2 + m_j^2)\, dt \le \frac{\alpha^2}{2\sigma^2}.$$

Coupling with the zero-effort scores and applying Cauchy–Schwarz yields $\mathcal{B} - b(T) \le \alpha\sqrt{T}/(\sqrt{2}\sigma)$. In the first precision region, (31) and $S_T \ge S_0$ bound both its linear and quadratic terms by a constant times α, uniformly for $T \le \bar{T}$. In the other region the competitive-output bound does the same, while the displayed full-disclosure estimate covers the endpoint $\lambda = \infty$. Consequently there is $C_\sigma < \infty$, independent of c, T, λ, such that, for all sufficiently small $\alpha > 0$,

$$\mathcal{B} - b(T) \le C_\sigma \alpha, \qquad 0 \le T \le \bar{T}, \quad 0 < \lambda \le \infty.$$

Because T_0 maximizes $u(b(T)) - D(T)$, concavity and $u'(b(T)) \le u'(b_0)$ give, for sufficiently large c,

$$u(\mathcal{B}) - c\alpha - D(T) \le V_0 + \alpha\,[u'(b_0)C_\sigma - c] < V_0.$$

No positive-prize design can therefore compete. The feasible design $(\theta, T) = (0, T_0)$ attains V_0. Moreover, the right derivative of $u(b(T)) - D(T)$ at zero equals $u'(b_0)\sigma^2/(2\sqrt{2\pi S_0}) > 0$, so $T_0 > 0$; all precision choices give this same payoff. This proves part (iii). □

Proof of Proposition 2 Hold the prescribed initial public input fixed under both parameter vectors, with the initial latent law in equation (10). Suppose two interior vectors Θ and Θ' induce the same joint distribution of submission counts and public readouts. Identification compares all possible data histories, not different observations along a single realized history.

Let E_ℓ be the event that all earlier counts are zero and $n_\ell^i + n_\ell^j > 0$, so the first readout occurs at s_ℓ. Before this readout, the recursion carries forward the fixed initial public input and the intensities are deterministic conditional on the parameters. Finite, positive intensities make each preceding zero-count probability and the current nonzero-total-count probability strictly positive. Their finite product gives $\Pr_\Theta(E_\ell) > 0$ and $\Pr_{\Theta'}(E_\ell) > 0$.

Conditional on all earlier counts being zero, contestant $k \in \{i, j\}$ has zero-count probability $\exp\{-\Delta_{\text{num}} \tau_k^{(1)}(s_{\ell-1})\}$ in interval ℓ. Equality of the joint laws implies equality of these conditional probabilities, so both intensity sequences coincide along the no-previous-submission history. Count draws along this history are independent of the initial latent uncertainty and innovation shocks; conditioning on E_ℓ therefore leaves their joint distribution unchanged.

To see the first implication explicitly, equality of the conditional zero-count probabilities and injectivity of the exponential map give $\tau_k^{(1)}(s_{\ell-1}; \Theta) = \tau_k^{(1)}(s_{\ell-1}; \Theta')$ recursively for each k and ℓ. The additional condition that the total count in interval ℓ is positive depends only on the Poisson draws at that interval. Because its intensity is already fixed along this history, it does not tilt the independent Gaussian state and measurement shocks.

Along this history, define the cumulative intensity difference

$$A(s_\ell) := \Delta_{\text{num}} \sum_{q=1}^{\ell} \left[\tau_i^{(1)}(s_{q-1}) - \tau_j^{(1)}(s_{q-1}) \right]. \tag{21}$$

The identified intensity sequences make $A(s_\ell)$ identical under Θ and Θ'. On E_ℓ, the accumulated drift is $A(s_\ell)/r$. The independent Gaussian initial uncertainty and innovation give latent variance $S_0 + \sigma^2 s_\ell$. Since $h_1 = s_\ell$, the fresh independent measurement error adds variance $1/(\lambda s_\ell)$. Thus

$$\hat{y}_{s_\ell} \mid E_\ell, \Theta \sim \mathcal{N}\left(\mu_0 + \frac{A(s_\ell)}{r}, S_0 + \sigma^2 s_\ell + \frac{1}{\lambda s_\ell} \right). \tag{22}$$

Since E_ℓ has positive probability under both vectors, equality of the joint laws implies equality of these conditional Gaussian distributions. No future submission times are conditioned on.

Equality of the first-readout variances under Θ and Θ', multiplied by s_ℓ, gives

$$(\sigma^2 - \sigma'^2) s_\ell^2 + (S_0 - S_0') s_\ell + \left(\lambda^{-1} - \lambda'^{-1} \right) = 0.$$

For $\ell = 1, 2, 3$, the three distinct positive times s_ℓ are roots of the same polynomial of degree at most two. The polynomial is therefore identically zero. Its constant, linear, and quadratic coefficients give $\lambda^{-1} = \lambda'^{-1}$, $S_0 = S_0'$, and $\sigma^2 = \sigma'^2$, respectively. Interior positivity then yields $\lambda = \lambda'$ and $\sigma = \sigma'$.

Equality of the first-readout means gives

$$\mu_0 + \frac{A(s_\ell)}{r} = \mu_0' + \frac{A(s_\ell)}{r'}.$$

At s_1, equation (14) has a common strictly positive kernel for the two contestants and differs only through $1/c_k$. Since $r, \theta > 0$ and $c_i \neq c_j$, this implies

$$A(s_2) - A(s_1) = \Delta_{\text{num}} \left[\tau_i^{(1)}(s_1) - \tau_j^{(1)}(s_1) \right] \neq 0.$$

Subtracting the mean equalities at these two times gives

$$[A(s_2) - A(s_1)]\left(r^{-1} - r'^{-1}\right) = 0,$$

so positivity of r, r' gives $r = r'$. Substitution into either mean equality then gives $\mu_0 = \mu_0'$. Finally, the identified initial intensities and $\tilde{y}_0 = \mu_0$ recover both costs directly:

$$c_k = \frac{r\theta}{\tau_k^{(1)}(0)\sqrt{2\pi(S_0 + \sigma^2 T)}} \exp\left\{-\frac{\mu_0^2}{2(S_0 + \sigma^2 T)}\right\}, \qquad k \in \{i, j\}.$$

Thus $\Theta = \Theta'$, establishing local identification. At $c_i = c_j$, the two leading-order efforts coincide, so their difference and hence $A(s_\ell)$ are zero along every public history. A common rescaling of (r, c_i, c_j) leaves each ratio r/c_k, and therefore both count intensities, unchanged; because the effort difference remains zero, it also leaves every readout prediction unchanged. The full vector is therefore not identified at equality. □

Appendix B: Auxiliary Results

B.1. Lemmas for Joint Design

This section establishes the equilibrium and output comparisons used in Proposition 1. We first record the full-disclosure benchmark and the output identities, then establish positive-precision equilibrium existence and compare precision at weak incentives. Finally, we bound competing prize–deadline choices to obtain the global payoff comparison supporting part (ii).

Full-disclosure benchmark. For full disclosure, specialize the strategies and values in Ryvkin (2022, Propositions 1 and 3, equations (18) and (23)) to $c_i = c_j = c$ and $V = \theta > 0$. Put $\alpha = \theta/c$ and $w = \alpha/\sigma^2$. The normalized values $U_k = V^k/\theta$ satisfy $U_i + U_j = f(d)$ and $U_i - U_j = d$, where

$$f(d) = 1 + \frac{2}{w}\log\frac{\cosh(wd/2)}{\cosh(w/2)}, \qquad K(d) = \frac{d + \sinh(wd)/w}{2}, \tag{23}$$

and $d(y,t)$ is determined by

$$K(d(y,t)) = K(1)\left[2\Phi\left(\frac{y}{\sigma\sqrt{T-t}}\right) - 1\right]. \tag{24}$$

The change of variables is $\rho_s = \tanh(wd/2)$ in Ryvkin's notation. Since the total prize is θ, summing the cited value functions gives

$$\mathbb{E}_{y,t}\int_t^T (m_i^2 + m_j^2)\, ds = 2\alpha[1 - f(d(y,t))]. \tag{25}$$

For finite precision, put $Q_t = S_t + \sigma^2(T-t)$ and $\beta(y,q) = \mathbb{E}|\mathcal{N}(y,q)|/2$. Since $\beta_y = \Phi(y/\sqrt{q}) - 1/2$, $\beta_q = \beta_{yy}/2$, and $Q_t' = -\lambda S_t^2$, the drift from $\beta_q Q_t'$ cancels the quadratic-variation term from $d\tilde{y}_t = (m_i - m_j)dt + \sqrt{\lambda} S_t\, d\widehat{B}_t$. Also, $\max\{x_T^i, x_T^j\} = (x_T^i + x_T^j)/2 + |y_T|/2$ and $\mathbb{E}[|y_T|/2 \mid I_T] = \beta(\tilde{y}_T, S_T)$. Applying Itô's formula to the mean score and to $\beta(\tilde{y}_t, Q_t)$, then using $\beta(0, Q_0) = \sqrt{(S_0 + \sigma^2 T)/(2\pi)}$, gives

$$\mathcal{B} = b(T) + \mathbb{E}\int_0^T \left\{\frac{m_i + m_j}{2} + (m_i - m_j)\left[\Phi(\tilde{y}_t/\sqrt{Q_t}) - \frac{1}{2}\right]\right\} dt, \qquad b(T) = \sqrt{\frac{S_0 + \sigma^2 T}{2\pi}}. \tag{26}$$

The Gaussian flux estimate used below is

$$G_q(y) - \int_0^q (|G'_{q-v}| * G_v^2)(y)\, dv = \frac{2}{\pi}\int_0^\infty \frac{G_q(\sqrt{2}|y|\cosh h)}{2\cosh^2 h - 1}\, dh \geq 0. \tag{27}$$

The identity follows by taking the Laplace transform in q.

LEMMA 1 **(Positive-precision equilibrium).** *For $\theta, T, \sigma, S_0, c_i, c_j > 0$ and $0 < \lambda < \infty$, the normalized HJB system has a unique bounded classical solution with bounded spatial gradients. Its feedback policies form an interior Markov-perfect equilibrium.*

Proof of Lemma 1 Use $U^k = V^k/\theta$, $a_t = \lambda S_t^2$, and $\alpha_k = \theta/c_k$ as in the proof of Theorem 1. For fixed positive finite precision, a_t is smooth and bounded above and away from zero on $[0,T]$. The terminal functions $g_i(y) = \Phi(y/\sqrt{S_T})$ and $g_j = 1 - g_i$ are smooth and bounded, with bounded derivatives.

We apply Theorem 2.7 of Jackson (2024) to $U_t^k + (a_t/2)U_{yy}^k + F_k(U_y) = 0$ on $[0,T] \times \mathbb{R}$. For $z = (z_i, z_j)$, the required structure follows from

$$F_k(z) = z_k(\alpha_i z_i + \alpha_j z_j) - \frac{\alpha_k}{2} z_k^2, \qquad k \in \{i, j\}.$$

Thus the common-drift term in the theorem's decomposition is $b_0(z) = \alpha_i z_i + \alpha_j z_j$, while $b^k(z) = -\alpha_k z_k^2/2$. The driver is smooth and satisfies $|F_k(z)| \le C|z_k||z|$, which verifies the quadratic-growth and local Lipschitz conditions. For the a-priori boundedness condition, take the coordinate vectors $\{\pm Le_i, \pm Le_j\}$ with $L \ge \max\{\alpha_i, \alpha_j\}$. They positively span $\mathbb{R}^2$. For $+Le_k$ the required upper bound is immediate because $b^k \le 0$; for $-Le_k$ it follows from

$$(-Le_k)^T b(z) = \frac{L\alpha_k}{2} z_k^2 \le \frac{L^2}{2} z_k^2 = \frac{1}{2}|(-Le_k)^T z|^2.$$

The scalar diffusion coefficient is uniformly elliptic and smooth in time, while F is time independent and the Gaussian terminal functions are $C^{2,\beta}$ for every $\beta \in (0,1)$. These observations verify the remaining regularity hypotheses of the cited classical-solution theorem. The cited theorem therefore gives a unique bounded classical solution with bounded spatial gradients.

Parabolic regularity gives bounded second spatial derivatives. As in the proof of Theorem 1, Feynman–Kac with terminal data $U_y^i = -U_y^j = G_{S_T} > 0$ gives $U_y^i > 0$ and $U_y^j < 0$. The policies $m_i = \alpha_i U_y^i$ and $m_j = -\alpha_j U_y^j$ therefore form an interior Markov-perfect equilibrium by (17). □

LEMMA 2 **(Full disclosure at weak incentives).** *Suppose $c_i = c_j = c > 0$. There is $w_* > 0$ such that, whenever $S_0/(\sigma^2 T) \in [99/2, 198]$ and $0 < \theta/(c\sigma^2) < w_*$, the selected symmetric equilibria satisfy*

$$\mathcal{B}(0, \theta, T, \infty; S_0) > \mathcal{B}(0, \theta, T, \lambda; S_0) \qquad \text{for every } 0 < \lambda < \infty. \tag{28}$$

Proof of Lemma 2 Measure time in units of T and scores in units of $\sigma\sqrt{T}$. Write $w = \theta/(c\sigma^2)$, $s = S_0/(\sigma^2 T)$, $\ell = \lambda\sigma^2 T^2$, and $q = 1 + s$. In these units, $S_t' = 1 - \ell S_t^2$ and $Q_t = S_t + 1 - t$. Denote output by $\widehat{\mathcal{B}}(w, s, \ell)$. All estimates below are uniform for $s \in [99/2, 198]$. Equation (26) gives the common first variation $G_q(0)$ at zero prize. For full disclosure, the explicit equilibrium gives the same coefficient:

$$\widehat{\mathcal{B}}(w, s, \infty) = \sqrt{q/(2\pi)} + wG_q(0) + O(w^2). \tag{29}$$

Bounded precisions. For every finite L, the positive lower bound on S_T makes the analytic construction in Theorem 1 uniform on $[99/2, 198] \times (0, L]$. The coefficients and expansion extend continuously to the limiting endpoint $\ell = 0$, which is used only to obtain a uniform comparison for arbitrarily small positive precision. Indeed, the analytic-norm proof uses only contraction of the Gaussian propagators and a uniform analytic

bound on the terminal density; it does not divide by the diffusion coefficient. Its contraction radius and remainder constant are therefore uniform for $\ell \in [0, L]$. The Riccati solution and every coefficient in the expansion depend continuously on (s, ℓ) on this compact set. The gradient mild equations and the output identity give, uniformly,

$$\widehat{\mathcal{B}}(w, s, \infty) - \widehat{\mathcal{B}}(w, s, \ell) = w^2 \Delta_2(s, \ell) + O(w^3).$$

The full-disclosure formulas (23)–(24) are analytic in w at zero. With $A_t = s + t - S_t$, Gaussian integration gives $\Delta_2(s, \ell) = b_2(s, \infty) - b_2(s, \ell)$, where

$$b_2(s, \ell) = \int_0^1 \int_t^1 K_q(A_t, A_r)\, dr\, dt, \qquad K_q(A, B) = \frac{A}{\pi\sqrt{2\pi}\sqrt{q - B}(q + B)\sqrt{q^2 + qB - 2A^2}}.$$

For $\ell > 0$, $\rho_t = A_t/(s + t)$ increases strictly from zero and stays below one: its derivative has the opposite sign to $E_t = (s + t)S_t' - S_t < 0$, since either $S_t' \le 0$ or $E_0 = -\ell s^3 < 0$ and $E_t' = -2\ell(s + t)S_t S_t' < 0$. The kernel is strictly increasing in its first argument because

$$\partial_A \log K_q(A, B) = \frac{1}{A} + \frac{2A}{q^2 + qB - 2A^2} > 0.$$

It also increases along radial rays: for $x = B/q$ and $z = A/q$,

$$\frac{AK_A + BK_B}{K_q} = \frac{1 + x^3 + xz^2 - 3x^2z^2}{(1 - x^2)(1 + x - 2z^2)} > 0, \qquad 0 \le z \le x < 1.$$

The numerator is affine in z^2, so its minimum is either $1 + x^3$ or $(1 - x)(3x^3 + x^2 + x + 1)$, both positive. Since $0 < \rho_t < \rho_r < 1$ for $0 < t < r < 1$, first increasing the first argument and then moving along the radial ray gives

$$K_q(A_t, A_r) < K_q(\rho_r(s + t), \rho_r(s + r)) < K_q(s + t, s + r).$$

At $\ell = 0$, the Riccati equation gives $S_t = s + t$, hence $A_t = 0$ and $b_2 = 0$. The endpoint bound $C[(1 - t)(1 - r)]^{-1/2}$ gives continuity by dominated convergence. For $\ell > 0$ the strict kernel inequality above gives $b_2(s, \ell) < b_2(s, \infty)$, while the same conclusion at $\ell = 0$ follows from positivity of the full-disclosure integral. Thus Δ_2 is positive and continuous on $[99/2, 198] \times [0, L]$, so it has a strictly positive minimum. Choosing w small enough for the uniform $O(w^3)$ remainder proves the comparison for $0 < \ell \le L$.

Unbounded precisions. Put $\eta = \ell^{-1/2} < 1/4$. Use an auxiliary contest with constant posterior variance η and initially disclosed mean variance $s - \eta$; total initial uncertainty remains s. We first identify a positive leading term in the full-disclosure output advantage. With $\tau = 1 - t$, set $h = \tau + \eta$, $v = \log(h/\eta)$, and $z = y/\sqrt{h}$. The full-disclosure profiles R_w, D_w, b_w satisfy

$$\mathcal{L}_0 R_w + w\{3(R_w')^2 - (D_w')^2\}/4 = 0, \qquad \mathcal{L}_0 D_w + (w/2)R_w' D_w' = 0,$$

$$(\mathcal{L}_0 - 1/2)b_w + wR_w' b_w' + (w/2)D_w' = 0, \qquad \mathcal{L}_0 = (\partial_{zz} + z\partial_z)/2.$$

Here $R_0 = 1$, $D_0 = 2\Phi - 1$, $b_0 = \beta = G_1 + z(\Phi - 1/2)$, and $R_w = 1 - w\Phi(1 - \Phi) + O(w^2)$ by the explicit equilibrium.

The profile error $E = (\phi, \psi, \chi)$ satisfies $E_v = \mathcal{A}_w E + \mathcal{N}_w(E)$, where $\mathcal{A}_w$ is the linearization, $\mathcal{N}_w$ the quadratic remainder, and $E(0) = (1 - R_w, 2\Phi - 1 - D_w, \beta - b_w)$. Explicitly,

$$\mathcal{N}_w(E) = w\left(\frac{3\phi'^2 - \psi'^2}{4}, \frac{\phi'\psi'}{2}, \phi'\chi'\right),$$

while the player block of $\mathcal{A}_w$ is

$$\begin{pmatrix} \mathcal{L}_0\phi + w(3R_w'\phi' - D_w'\psi')/2 \\ \mathcal{L}_0\psi + w(D_w'\phi' + R_w'\psi')/2 \end{pmatrix},$$

and its organizer block is

$$(\mathcal{L}_0 - 1/2 + wR_w'\partial_z)\chi + w(\phi' b_w' + \psi'/2).$$

The weighted-space statement concerns the perturbations, not the linearly growing background profiles β and b_w separately. After subtracting the $w=0$ profile, the stationary profile equations show that $1-R_w$, $2\Phi - 1 - D_w$, and $\beta - b_w$, together with their required derivatives, have Gaussian decay. Consequently $E(0)$ belongs to the fractional-domain space used below, even though neither β nor b_w belongs to the weighted space by itself. On the even–odd–even subspace of $\mathcal{H} = L^2(e^{z^2/2}dz)^3$, put $X = D((-\mathcal{A}_0)^{7/8})$. At $w=0$ these blocks are Ornstein–Uhlenbeck operators. Their Hermite spectrum has the simple leading eigenpair $(-1/2, (G_1, 0, 0))$; parity places every other eigenvalue at or below -1. Moreover,

$$\|(\mathcal{A}_w - \mathcal{A}_0)f\|_{\mathcal{H}} \le C|w|\|(-\mathcal{A}_0)^{1/2}f\|_{\mathcal{H}}.$$

Spectral perturbation (Kato 1980, Chapter VII, Sections 1–2) and sectorial estimates (Lorenzi et al. 2005, Proposition 3.2.2 and Chapter 5) therefore preserve one simple eigenvalue $\nu(w)$ with $|\nu(w) + 1/2| < 1/20$ and give decay rate $7/10$ on the complementary subspace. If Π_w is its spectral projection, the corresponding uniform semigroup estimates are

$$\|e^{v\mathcal{A}_w}\|_{\mathcal{H}\to X} \le C(1+v^{-7/8})e^{-9v/20}, \qquad \|e^{v\mathcal{A}_w}(I-\Pi_w)\|_{\mathcal{H}\to X} \le C(1+v^{-7/8})e^{-7v/10}.$$

Sobolev embedding gives $\|\mathcal{N}_w(E)\|_{\mathcal{H}} \le C|w|\|E\|_X^2$; since $7/8 < 1$, the singular factor is integrable in Duhamel's formula. The resulting map is a contraction in $\sup_{v\ge 0} e^{2v/5}\|E(v)\|_X$ when w is small. Thus linearized stability (Lorenzi et al. 2005, Theorem 7.1.3) gives $\|E(v)\|_X \le C|w|e^{-2v/5}$, uniformly on a small complex prize disc.

Normalize the slow right eigenvector e_w and its left functional ℓ_w by $\ell_w(e_w) = 1$. Projection of Duhamel's formula gives the exact amplitude

$$a(w) = \ell_w E(0) + \int_0^\infty e^{-\nu(w)r}\ell_w \mathcal{N}_w(E(r))\,dr.$$

The integral is $O(w^3)$ because $C|w|^3 e^{-(4/5 + \mathrm{Re}\,\nu(w))r}$ is integrable, whereas $R_w = 1 - w\Phi(1-\Phi) + O(w^2)$ and $\int \Phi(1-\Phi)dz = 1/\sqrt{\pi}$ give $\ell_w E(0) = w/\sqrt{\pi} + O(w^2)$. Hence $a(w) = w/\sqrt{\pi} + O(w^2) > 0$ for small real $w > 0$. The designer component ξ_w of the eigenvector, normalized to $(G_1, 0, 0)$ at $w=0$, is $-wG_1\beta + O(w^2)$: its first-order equation is $-\mathcal{L}_0\xi_1 = G_1'(\Phi - 1/2)$, verified directly by $\mathcal{L}_0(G_1\beta) = G_1'(\Phi - 1/2)$. The fast designer component at order w is exactly $-we^{-v}G_1$. Spectral decay and Cauchy's estimate therefore give

$$\chi(v) = a(w)e^{\nu(w)v}\xi_w - we^{-v}G_1 + O_X(w^2 e^{-7v/10}).$$

At the initial date, $e^v = (1+\eta)/\eta$. Since $\nu(w) = -1/2 + O(w)$, the slow mode is of order $w^2\eta^{1/2+O(w)}$; the exponent $11/20$ below leaves a uniform slack for sufficiently small w. More precisely, at the initial date $h = 1+\eta$ and $Y_h \sim \mathcal{N}(0, (s-\eta)/h)$. The auxiliary output is $\sqrt{h}\,\mathbb{E}[b_w(Y_h) + \chi(v, Y_h)]$. Its background difference from full disclosure is $w\eta G_q(0) + O(w^2\eta)$, which cancels $-\sqrt{h}\,we^{-v}\mathbb{E}G_1(Y_h) = -w\eta G_q(0)$. Since $\mathbb{E}[G_1(Y_h)\beta(Y_h)]$ has a positive uniform lower bound, constants $d, C > 0$ give

$$\widehat{\mathcal{B}}(w, s, \infty) - \widehat{\mathcal{B}}_{\mathrm{aux}}(w, s, \eta) \ge w^2\{d\eta^{11/20} - C(\eta^{7/10} + \eta)\}.$$

It remains to compare the auxiliary contest with the original path. In the common variance clock Q, their incentive weights are $\gamma_o = \eta^2/S_{t(Q)}^2$ on $[S_1, q]$, zero below S_1, and $\gamma_a = \mathbf{1}_{[\eta,1+\eta]}$. Both lie in $[0,1]$, have integral one, and the Riccati solution gives $\int |\gamma_o - \gamma_a|\, dQ < 2\eta$. More precisely, $S_t = \eta(1+re^{-2t/\eta})/(1-re^{-2t/\eta})$, where $r = (s-\eta)/(s+\eta) \in (0,1)$. Thus $\varepsilon_\eta := S_1 - \eta \le C\eta e^{-2/\eta}$, and $|\gamma_o - \gamma_a| \le 4e^{-1/\eta}$ below $Q = 1/2$ except on $[\eta, S_1]$.

Extend both systems below their terminal variances by zero incentive weight. Their gradients start from the common heat profile $p = n = G_Q$; in the mild equations, w is replaced by $w\gamma(Q)$. The Gaussian bilinear bound in (27) makes these equations contractions for sufficiently small $w > 0$, uniformly in η. For conditional output J, set $Z = \partial_y(J - \beta)$, where $\beta(y,Q) = \mathbb{E}|\mathcal{N}(y,Q)|/2$. Its mild equation is obtained from

$$J_Q = \tfrac{1}{2} J_{yy} + w\gamma\{(p-n)J_y + (p+n)/2\}.$$

The same bound and Gaussian smoothing give $|p - G_Q| + |n - G_Q| \le CwG_Q$, $\|Z\|_\infty \le Cw$, and bounded spatial derivatives on $1/2 \le Q \le q$. Let $\mathfrak{d}$ be the sum of the two gradient differences in the norm $\sup_{Q,y} |f(Q,y)|/G_Q(y)$ and the difference of Z in the supremum norm. Subtract the three mild equations. Above $Q = 1/2$, differentiation of the bounded source profiles bounds the weight-error terms by $C|w|\eta$. Below that level, the bilinear bound controls the uniform weight error. On the exceptional interval, Gaussian multiplication gives

$$\frac{(|G'_{Q-v}| * G_v^2)(y)}{G_Q(y)} \le C\left\{\frac{1}{v} + \frac{1}{\sqrt{v(Q-v)}}\right\}, \qquad 0 < v < Q.$$

Integration over an interval of length ε_η with $v \ge \eta$ bounds its contribution by $C|w|\sqrt{\varepsilon_\eta/\eta}$. The same bound for the output gradient follows from $\|\partial_y P_v\|_{L^\infty \to L^\infty} \le Cv^{-1/2}$. The remaining terms are Lipschitz in the three unknowns, so

$$\mathfrak{d} \le C|w|\{\eta + e^{-1/\eta} + \sqrt{\varepsilon_\eta/\eta}\} + C|w|\mathfrak{d} \le C|w|\eta + C|w|\mathfrak{d}.$$

Take $Cw < 1/2$, so $\mathfrak{d} \le Cw\eta$. To compare output, remove the common first-order source G_Q: $\mathcal{R}_\gamma = (p+n)/2 - G_Q + (p-n)(\beta_y + Z)$ satisfies $|\mathcal{R}_\gamma| \le CwG_Q$ and $|\mathcal{R}_o - \mathcal{R}_a| \le C\mathfrak{d}G_Q$. With P_v the heat semigroup, $P_{q-Q}G_Q = G_q$ and $\int_0^q \gamma\, dQ = 1$ give the exact root identity

$$J_\gamma(q,0) = \beta(0,q) + wG_q(0) + w\int_0^q \gamma(Q)[P_{q-Q}\mathcal{R}_\gamma(Q)](0)\, dQ.$$

The common first-order term cancels, hence

$$|J_o(q,0) - J_a(q,0)| \le Cw\left(w\int_0^q |\gamma_o - \gamma_a|\, dQ + \mathfrak{d}\right) \le Cw^2\eta.$$

Because $11/20 < 7/10 < 1$, the positive term decays more slowly than both error terms. Choose $\eta_* > 0$ so that $d\eta^{11/20} > C(\eta^{7/10} + \eta)$ for $0 < \eta < \eta_*$. Combine this tail bound with the compact comparison for $L = \eta_*^{-2}$ and take the smaller prize threshold. Rescaling proves (28). □

LEMMA 3 **(A full-disclosure payoff comparison)**. *Suppose $c_i = c_j = c$, u satisfies the curvature condition in Proposition 1, and S_0 is sufficiently large. There exist $\sigma > 1$ and $0 < c < c_e$, where $T_0 > 0$ maximizes $u(\sqrt{(S_0 + \sigma^2 T)/(2\pi)}) - D(T)$ and*

$$c_e = \frac{T_0 u'(\sqrt{(S_0 + \sigma^2 T_0)/(2\pi)})}{\sqrt{2\pi(S_0 + \sigma^2 T_0)}}.$$

Let V_∞ be the supremum of the organizer's payoff over full-disclosure designs at these parameters. Every positive-prize, positive-deadline finite-feedback design has payoff strictly below V_∞.

Proof of Lemma 3 The utility condition gives $-Bu''(B)/u'(B) \geq \zeta B \to \infty$ and $Bu'(B) \to 0$. We construct a parameter point at which only weak incentives can compete with the zero-prize design, then apply Lemma 2.

Put $k = 1/100$, $b = \sqrt{S_0/\{2\pi(1-k)\}}$, and $M = bu'(b)$. Choose

$$T_0 D'(T_0) = Mk/2, \qquad \sigma^2 T_0 = kS_0/(1-k).$$

The second equality makes $b(T_0) = b$. It also gives

$$T_0 u'(b) b'(T_0) = \frac{Mk}{2} = T_0 D'(T_0),$$

so T_0 satisfies the zero-prize first-order condition. Because $u(b(T)) - D(T)$ is strictly concave, this condition makes T_0 its unique maximizer. As $S_0 \to \infty$, $M \to 0$ by the utility-tail assumption. The first equality and $D''(0) > 0$ then imply $T_0 \to 0$, while the second implies $\sigma^2 = kS_0/\{(1-k)T_0\} \to \infty$. Write $V_0 = u(b) - D(T_0)$, $\alpha = \theta/c$, $w = \alpha/\sigma^2$, and $x = T/T_0$. Then $c_e\sigma^2 = Mk$. Monotonicity of $D'(T)/T$ gives

$$D(T) - D(T_0) \geq Mk(x^2-1)/4, \qquad D(T_0) \leq Mk/4.$$

Indeed, $D'(s) \geq (s/T_0)D'(T_0)$ for $s \geq T_0$ and the reverse inequality holds for $s \leq T_0$; integrating on the appropriate interval gives the first bound. Integrating $D'(s) \leq (s/T_0)D'(T_0)$ on $[0, T_0]$ gives the second.

We choose a sufficiently large fixed γ below, then increase S_0 so that $-Bu''(B) \geq (\gamma+1)u'(B)$ for $B \geq .99b$ and $D'(T_0)/\{D''(0)T_0\} < 1 + 10^{-5}$. At $c = c_e$, any design with payoff at least V_0 satisfies the budget $\theta + D(T) \leq u_\infty - V_0 \leq M/\gamma + Mk/4$, where $u_\infty = \lim_{B\to\infty} u(B)$ and the tail bound follows by integrating the curvature inequality above b: it gives $u'(B) \leq u'(b)(B/b)^{-(\gamma+1)}$ and hence $u_\infty - u(b) \leq bu'(b)/\gamma = M/\gamma$. Since $D(T) \geq D''(0)T^2/2$, this gives

$$w + \frac{D''(0)T_0}{D'(T_0)}\frac{x^2}{4} \leq \frac{1}{4} + \frac{1}{\gamma k}.$$

Take $\gamma \geq 10^6$; this implies $w + x^2/4 < .252$, hence $x < 2$. Moreover, concavity gives $u(b) - u(.99b) \geq .01M > D(T_0)$, so a design with $\mathcal{B} \leq .99b$ has payoff strictly below $u(b) - D(T_0) = V_0$. Competitiveness therefore requires $\mathcal{B} > .99b$.

A coarse output bound. We verify, for this entire competitive set,

$$\mathcal{B} \leq b(T) + \alpha T G_{S_0+\sigma^2 T}(0)(1 + 2.9w), \qquad b(T) = \sqrt{(S_0 + \sigma^2 T)/(2\pi)}. \tag{30}$$

Put $Q_t = S_t + \sigma^2(T-t)$, $a_t = \lambda S_t^2$, $U_i = V_i/\theta$, $U_j = V_j/\theta$, $p = \partial_y U_i$, $n = -\partial_y U_j$, $H = p + n$, and $A = p - n$.

If $\lambda S_0^2 < .72\sigma^2$, the Riccati path increases and $S_T \geq S_0$. Thus

$$e := \alpha T/S_T \leq 2w/99 < .01, \qquad Q_0/S_T \leq 101/99.$$

Here we used $x < 2$ and $S_0/(\sigma^2 T_0) = (1-k)/k = 99$, so $\sigma^2 T/S_0 < 2/99$; the two displayed bounds then follow from $S_T \geq S_0$. Normalize scores by $\sqrt{S_T}$ and reverse time by $v = \alpha(T-t)/S_T$. In the Fourier-moment scale of Theorem 1, the normalized terminal Gaussian has the coefficientwise majorant

$$G_1(0)e^{r^2/2} + \frac{1}{\pi}\sum_{j\geq 0}\frac{r^{2j+1}}{(2j+1)!!} \leq \Psi(r) := .4e^{r^2/2} + .32r/(1 - r^2/3), \qquad 0 \leq r < \sqrt{3}.$$

Put $\kappa = 7/4$. For nonnegative moment majorants X, Y of H, A, respectively, the weighted sum of their quadratic fluxes satisfies

$$\frac{XY}{4} + \frac{\kappa}{8}(3Y^2 + X^2) \le \frac{\kappa}{8}(X + \kappa Y)^2,$$

because $\kappa^2 > 3$. Heat contraction and Picard iteration therefore majorize the weighted moment sum $(X + \kappa Y)/2$ by the scalar equation $h_v = \kappa h h_r$, $h(0, r) = \Psi(r)$, whose analytic branch satisfies $h(v, r) = \Psi(r + \kappa v h(v, r))$. In the Picard expansion, subtracting the propagated Gaussian removes the order-zero term. Each individual gradient correction is bounded by half the sum of the moment majorants of $H - 2G_Q$ and A, so its Fourier norm is at most $S_T^{-1/2}\{h(v, 0) - .4\}$. Direct substitution gives $h(v, 0) - .4 \le v/4$ for $v \le .01$: at the largest argument $r = (7/4)(.01)(.4025)$,

$$\frac{.4(r^2/2)e^{r^2/2} + .32r/(1 - r^2/3)}{.01} < \frac{1}{4}.$$

Thus $d(t) = \alpha(T - t)/(4S_T^{3/2})$ bounds each correction. Since the Fourier-moment norm dominates the pointwise norm, this means

$$|p - G_{Q_t}| \le d(t), \qquad |n - G_{Q_t}| \le d(t),$$

and hence, with $E = H - 2G_{Q_t}$, $|E| \le 2d(t)$ and $|A| \le 2d(t)$. These are pointwise bounds; the Gaussian-weighted supremum norm used in the next, high-diffusion case is a different norm.

Let $J(t, y)$ be the conditional continuation contribution to $\mathcal{B} - b(T)$. It has terminal value zero and solves

$$J_t + \mathscr{L}_t J + \frac{\alpha}{2}\{H + [2\Phi(y/\sqrt{Q_t}) - 1]A\} = 0, \qquad \mathscr{L}_t = \frac{a_t}{2}\partial_{yy} + \alpha A \partial_y.$$

Consider the unnormalized barrier

$$\overline{J}(t, y) = \alpha(T - t)G_{Q_t}(y) + \int_t^T 2\alpha d(s)\left\{1 + .486\frac{\alpha(T - s)}{Q_s}\right\} ds.$$

It has zero terminal value. Since $\partial_t G_{Q_t} + (a_t/2)\partial_{yy} G_{Q_t} = 0$,

$$(\partial_t + \mathscr{L}_t)\overline{J} = -\alpha G_{Q_t} + \alpha^2 A(T - t)G'_{Q_t} - 2\alpha d(t)\left\{1 + .486\frac{\alpha(T - t)}{Q_t}\right\}.$$

The source in the equation for J is $\alpha G_{Q_t} + \alpha E/2 + \alpha[2\Phi(y/\sqrt{Q_t}) - 1]A/2$. Using $|E|, |A| \le 2d$, $|2\Phi - 1| \le 1$, and $|G'_Q| < .243/Q$, the supersolution residual is at most

$$.486\alpha^2 d(t)\frac{T - t}{Q_t} + 2\alpha d(t) - 2\alpha d(t)\left\{1 + .486\frac{\alpha(T - t)}{Q_t}\right\} = -.486\alpha^2 d(t)\frac{T - t}{Q_t} \le 0.$$

Itô verification gives the shared low-diffusion bound

$$\mathcal{B} - b(T) \le \alpha T G_{S_0 + \sigma^2 T}(0) + \frac{\alpha^2 T^2}{4S_T^{3/2}}\left(1 + .486\frac{\alpha T}{S_T}\right). \tag{31}$$

This bound requires only $\lambda S_0^2 < .72\sigma^2$ and $\alpha T/S_T \le .01$. On the present competitive set, the excess output coefficient is $\sqrt{2\pi Q_0/S_T}(e/w)(.5 + .243e)/2 < .02$, proving (30) in this case.

If $\lambda S_0^2 \ge .72\sigma^2$, the whole Riccati path has $a_t \ge .72\sigma^2$. Indeed, $a'_t = 2\lambda S_t(\sigma^2 - a_t)$, so a_t moves monotonically toward σ^2 and never falls below $\min\{a_0, \sigma^2\} \ge .72\sigma^2$. In the norm $\|f\| = \sup|f|/G_{Q_t}$, write $E = H - 2G_Q$ and $\mathcal{T}(f, g) = \int_t^T \partial_y P_{Q_t - Q_s}(f_s g_s)\, ds$. The centered gradient equations are

$$E = \tfrac{\alpha}{2}\mathcal{T}(A, 2G_Q + E), \qquad A = A_1 + \tfrac{\alpha}{4}\{3\mathcal{T}(A, A) - \mathcal{T}(E, 4G_Q + E)\}.$$

Changing variables from time to Q_s gives $\alpha\|\mathcal{T}(f,g)\| \le K\|f\|\|g\|$, where $K = \alpha/\min a_t \le w/.72$. Indeed, (27) bounds the convolution integral after this variance substitution. The explicit source $A_1 = -\alpha(2\sqrt{\pi})^{-1}\int_t^T Q_s^{-1/2} G'_{Q_t - Q_s/2}\, ds$ has norm at most K: the same variance substitution evaluates its upper bound as $K|2\Phi(y/\sqrt{Q_t}) - 1|G_{Q_t}$. The centered map contracts on $\|E\| \le 6w^2$, $\|A\| \le 3w$. Indeed, for $0 < w \le .252$, its map-to-radius ratios satisfy

$$\max\left\{\frac{2+6w^2}{4\cdot.72}, \frac{1+w^2(51+36w^2)/4}{3\cdot.72}\right\} < 1,$$

and its Lipschitz row sums are bounded by

$$\frac{w}{.72}\max\left\{\frac{2+3w+6w^2}{2}, \frac{4+18w+12w^2}{4}\right\} < 1.$$

Lemma 1 identifies the fixed point with the equilibrium.

For its transition operator $\mathcal{K}_{t,s}$,

$$\mathcal{K}_{t,s} g \le \cosh^2(.35/2) P_{Q_t - Q_s} g, \qquad g \ge 0.$$

To justify the factor, note first that each player can choose zero effort, so $U_i, U_j \ge 0$, while summing their equilibrium payoffs gives the unit total prize minus nonnegative effort costs and hence $R := U_i + U_j \le 1$. Applying Itô's formula shows that e^{-KU_i} and e^{-KU_j} are submartingales. Their terminal values are exponentials of complementary probabilities in $[0,1]$, which gives $0 \le 1 - R \le 2\log\cosh(K/2)/K$. With $\kappa_t = \alpha/a_t$, Girsanov's log likelihood is

$$\kappa_s(R_s - 1) - \kappa_t(R_t - 1) - \int_t^s \kappa'_r(R_r - 1)\, dr - \frac{1}{4}\int_t^s \alpha\kappa_r(H_r^2 - A_r^2)\, dr, \qquad R = U_i + U_j.$$

Monotonicity of κ, in either direction, bounds this by $2\log\cosh(K/2)$; $H^2 - A^2 = 4pn \ge 0$ makes the last term nonpositive. Exponentiating and applying the heat semigroup gives the stated $\cosh^2(K/2)$ domination. Since $K \le w/.72 \le .35$, the displayed constant follows. Bounded gradients and positive diffusion justify the change of measure.

Let $\beta(y,Q) = \mathbb{E}|\mathcal{N}(y,Q)|/2$. Subtracting $\beta + \alpha(T-t)G_{Q_t}$ from conditional best output gives a zero-terminal linear equation with source $\alpha E/2 + \alpha A\beta_y + \alpha^2(T-t)AG'_{Q_t}$. Using $|\beta_y| \le 1/2$, $(T-t)/Q_t \le 1/\sigma^2$, and $|G'_Q| < .243/Q$, its absolute value is at most $\alpha w G_{Q_t}\{3w + 3/2 + .729w\}$. Since $\cosh^2(.35/2)\{3w + 3/2 + .729w\} < 2.9$ for $w \le .252$, the transition bound proves (30). This high-diffusion bound requires only $\lambda S_0^2 \ge .72\sigma^2$ and $w \le .252$, not the prior ratio.

Exclude prizes outside the weak-incentive range. Put $h = 1 + k(x-1)$ and $r = \sqrt{h} + kx(w + 2.9w^2)/\sqrt{h}$. Integrating the utility-curvature inequality gives $u(\mathcal{B}) - u(b) \le (M/\gamma)\{1 - (b/\mathcal{B})^\gamma\}$. The construction gives $b(T)/b = \sqrt{h}$, while

$$\frac{\alpha T G_{S_0 + \sigma^2 T}(0)}{b} = \frac{kxw}{\sqrt{h}}.$$

Thus (30) implies $\mathcal{B}/b \le r$. Since the curvature bound is increasing in $\mathcal{B}$, and since $\theta = c_e\alpha = Mkw$, substitution together with the delay bound yields

$$\frac{u(\mathcal{B}) - \theta - D(T) - V_0}{Mk} \le F_\gamma(x,w) := \frac{1 - r^{-\gamma}}{\gamma k} - w - \frac{x^2 - 1}{4}. \tag{32}$$

Let $w_* > 0$ be the threshold in Lemma 2 and put $w_0 = \min\{w_*/2, .252\}$. We need $F_\gamma < 0$ only on the compact set

$$\{(x,w) \in [0,2] \times [0,.252] : x \le 1/2 \text{ or } w \ge w_0\},$$

which excludes $(1,0)$.

On $w + (x^2-1)/4 \le 0$, we have $x \le 1$ and $w \le (1-x^2)/4$. Since r increases in w, $r < 1$ reduces to

$$x\left\{\frac{1+x}{4} + \frac{29(1-x)(1+x)^2}{160}\right\} < \frac{\sqrt{1-k(1-x)}}{1+\sqrt{1-k(1-x)}}, \qquad x < 1.$$

The left side is at most $.5 - (1-x)/40$, because the difference is $(1-x)^2(29x^2+87x+76)/160$. The right side exceeds this bound: its difference from $1/2$ is $k(1-x)/\{2(1+\sqrt{1-k(1-x)})^2\} < (1-x)/40$. Equality occurs only at $(1,0)$. The inequality $e^z \ge 1+z$ makes F_γ nonincreasing in γ. Every point of the displayed compact set eventually has $F_\gamma < 0$: if $w + (x^2-1)/4 \le 0$, then $r < 1$ and $F_\gamma \to -\infty$; otherwise $\limsup_{\gamma\to\infty} F_\gamma < 0$. The sets $\{F_\gamma < 0\}$ therefore increase with γ and cover the displayed compact set. A finite subcover gives one $\gamma \ge 10^6$ and a strictly negative uniform margin. Fix it before increasing S_0.

Finally take $c < c_e$ sufficiently close to c_e. The competitive budget still implies $w + x^2/4 < .252$, and replacing c_e by c changes the normalized prize term from $-w$ to $-(c/c_e)w$. Thus the right side of (32) gains only $(1-c/c_e)w$. The uniform margin excludes $x \le 1/2$ or $w \ge w_0$ after this small cost perturbation. Every competitive finite-feedback design therefore has $T_0/2 < T < 2T_0$ and $0 < w < w_*$. Lemma 2 strictly improves its payoff by full disclosure. All other designs have payoff below $V_0 \le V_\infty$. □

B.2. Sequential Leaderboard Likelihood

For the retained readouts defined in Section 3, let

$$g_k = \int_{t_{k-1}}^{t_k} \left[m_i^{(1)}(\tilde{y}_s, S_s, s) - m_j^{(1)}(\tilde{y}_s, S_s, s)\right] ds,$$

which is determined by the preceding public history, t_k, and Θ under the maintained carried-forward path. Set $R_k = (\lambda h_k)^{-1}$ and initialize $\mu_{0|0} = \mu_0$ and $P_{0|0} = S_0$. Conditional on $\mathcal{F}_{k-1}$, t_k, and Θ, prediction and measurement updating give

$$\mu_{k|k-1} = \mu_{k-1|k-1} + g_k, \qquad P_{k|k-1} = P_{k-1|k-1} + \sigma^2 h_k,$$
$$V_{k|k-1} = P_{k|k-1} + R_k, \qquad K_k = P_{k|k-1}/V_{k|k-1},$$
$$\mu_{k|k} = \mu_{k|k-1} + K_k(\hat{y}_{t_k} - \mu_{k|k-1}), \qquad P_{k|k} = (1-K_k)P_{k|k-1}.$$

Thus $\mu_{k|k-1}$ and $V_{k|k-1}$ are the predictive moments entering equation (13). After its density contribution is evaluated, $(\mu_{k|k}, P_{k|k})$ initializes the next prediction.

B.3. Solve S_t in Equation (4)

For $\lambda > 0$, put $\bar{S} = \sigma/\sqrt{\lambda}$ and $u_t = S_t/\bar{S}$. Equation (4) becomes

$$\dot{u}_t = \sigma\sqrt{\lambda}\,(1-u_t^2).$$

The constant solution is $u_t = 1$. Otherwise, separation of variables with the initial condition imposed directly gives

$$\int_{u_0}^{u_t} \frac{dv}{1-v^2} = \sigma\sqrt{\lambda}\,t.$$

If $u_0 < 1$, the antiderivative is $\tanh^{-1}$; if $u_0 > 1$, it is $\coth^{-1}$. Inverting the corresponding identity yields

$$S_t = \begin{cases} \bar{S} \cdot \tanh\left\{t \cdot \sigma\sqrt{\lambda} + \tanh^{-1}\left(S_0/\bar{S}\right)\right\} & \text{if } S_0 < \bar{S} \\ \bar{S} & \text{if } S_0 = \bar{S} \\ \bar{S} \cdot \coth\left\{t \cdot \sigma\sqrt{\lambda} + \coth^{-1}\left(S_0/\bar{S}\right)\right\} & \text{if } S_0 > \bar{S} \end{cases}$$

The vector field points toward $\bar{S}$, so the solution remains in the branch determined by S_0 and converges monotonically to $\bar{S}$. The steady-state variance diverges as $\lambda \downarrow 0$.

B.4. Consistency Benchmark for σ^2

This appendix derives the private-leaderboard benchmark for σ used in Section 5.2. For a given contest, let $\Delta y_k = y_k - y_{k-1}$ denote the private-leaderboard gap increment between two adjacent update times, and let $h_k = t_k - t_{k-1}$ be the corresponding elapsed time. Effort paths evaluated at the posterior mean parameters provide the plug-in drift adjustment $\hat{\mu}_k = \int_{t_{k-1}}^{t_k} [\hat{m}_i(s) - \hat{m}_j(s)]\, ds$. As a working Gaussian approximation, we use $\Delta y_k - \hat{\mu}_k \sim \mathcal{N}(0, \sigma^2 h_k)$, $k = 1, \ldots, n$. This is not an exact conditional distribution given an endogenous effort path. Treating these increments as approximately independent and holding the plug-in drift fixed, the working log-likelihood for σ^2 is, up to constants,

$$\ell(\sigma^2) = -\frac{n}{2}\log(\sigma^2) - \frac{1}{2}\sum_{k=1}^{n}\log h_k - \frac{1}{2\sigma^2}\sum_{k=1}^{n}\frac{(\Delta y_k - \hat{\mu}_k)^2}{h_k}.$$

Differentiating with respect to σ^2 and setting the derivative equal to zero gives

$$-\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2}\sum_{k=1}^{n}\frac{(\Delta y_k - \hat{\mu}_k)^2}{h_k} = 0.$$

Therefore,

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{k=1}^{n}\frac{(\Delta y_k - \hat{\mu}_k)^2}{h_k}. \tag{33}$$

In the empirical application, $\hat{\mu}_k$ is computed from effort paths evaluated at the posterior mean parameters, so this benchmark is a plug-in private-leaderboard consistency check rather than a fully model-free estimator. Moreover, asynchronous update timing means that adjacent displayed gaps may combine a new score for one contestant with a carried-forward score for the other. The expression in (33) is therefore an approximate benchmark for the carried-forward gap process, not an exact maximum likelihood estimator for synchronously observed latent diffusion increments.

Appendix C: Statistics

Table 8 reports contest-level estimates and diagnostic benchmarks; large λ^{bench} values can result from very small public–private score discrepancies. "Public Data %" reproduces Meta-Kaggle's reported public-leaderboard test-data share; it is not used in estimation or sample selection, and the recorded zero for contest 3288 does not indicate missing public-score observations.

Data construction. Focal-pair selection searches the top 20 contestants in the reconstructed final private standings. Contestants are ordered by their final carried-forward private scores; ties are broken by the order of the tied contestants' final score updates, with the later update ranked first. It first selects the highest-ranked contestant satisfying the individual activity criterion, then searches lower ranks for the first eligible

opponent whose activity window overlaps by more than ten days. A pair is retained only if both contestants have at least five submissions in the constructed observation window. For Don't Overfit! (contest 2489), θ is the \$250 prize for the prediction task; the separate variable-identification prize is excluded. The common score transformation in Section 5.1 does not make the different scoring metrics equivalent probabilities; each retains its substantive interpretation.

Within each submission date, preprocessing draws distinct second offsets with seed 1234 and assigns them in submission-ID order, retaining multiple submissions by the same contestant on the same date. Backfilling the earliest joint public gap supplies a data-dependent initial public input, not additional leaderboard observations. The fixed-input identification result in Proposition 2 does not cover the sampling law induced by this initialization. For eight contests preselected at evenly spaced ranks of submission counts, two alternative seeds give median absolute changes in the posterior mean of λ of 10.7% (full sample) and 13.4% (early sample); the largest shift is 2.47 baseline posterior standard deviations. All corresponding 95% posterior intervals overlap.

Regression inference. All cross-contest regressions in Section 5.2 use posterior mean parameter estimates, with HC1 heteroskedasticity-robust standard errors and confidence intervals. Reported inference does not propagate uncertainty from parameter estimation, focal-pair selection, or intraday time imputation.

Approximation diagnostics. In the full-sample fits, posterior medians of $w_k = \theta/(\sigma^2 c_k)$ are below one for 73 of the 146 player–contest pairs, with a cross-pair median of 1.000. The posterior median of $\max\{w_i, w_j\}$ is below one in 34 of 73 contests. These are descriptive summaries, not theoretical criteria for approximation accuracy.

C.1. Precision Counterfactual Calculations

For each accepted full-sample posterior draw s, the calibration uses the initial gap $\mu_{0,\ell s}$, innovation risk $\sigma_{\ell s}$, and symmetric cost $c_{\ell s} = 2/(c_{i,\ell s}^{-1} + c_{j,\ell s}^{-1})$. The latter preserves the leading-order coefficient on total effort. At the observed prize and duration, calibrated output is

$$\mathcal{B}_{\ell s}^{\mathrm{cal}}(\theta, T, \lambda) = \beta(\mu_{0,\ell s}, \sigma_{\ell s}^2 T) + \frac{\theta T}{c_{\ell s}\sqrt{2\pi Q_{\ell s}(\lambda)}} \exp\left[-\frac{\mu_{0,\ell s}^2}{2Q_{\ell s}(\lambda)}\right], \qquad Q_{\ell s}(\lambda) = \sigma_{\ell s}^2 T + \frac{\sigma_{\ell s}}{\sqrt{\lambda}},$$

where $\beta(\mu, q) = \mathbb{E}|Y|/2$ for $Y \sim \mathcal{N}(\mu, q)$. The no-effort term is held fixed across precisions; the incentive term uses steady-state uncertainty. Endpoint values are defined by limits. This symmetric calibration is not an exact asymmetric-equilibrium calculation.

We maximize the average output over all accepted draws to select one precision λ_ℓ^* per contest. Computation uses $x = 1/\sqrt{\lambda}$, so full disclosure is the endpoint $x = 0$. For each draw, the incentive derivative is negative once $x > (\mu_{0,\ell s}^2 - \sigma_{\ell s}^2 T)/\sigma_{\ell s}$, giving a finite search interval. We compare the endpoint with stationary maxima located on a logarithmic search grid and check stability after doubling grid resolution.

Panel A separately quantifies directional uncertainty by comparing each draw's optimal precision with its current precision $\lambda_{\ell s}$. Panel B uses the single policy λ_ℓ^* and reports

$$100\left[\frac{\sum_s \mathcal{B}_{\ell s}^{\mathrm{cal}}(\theta_\ell^{obs}, T_\ell^{obs}, \lambda_\ell^*)}{\sum_s \mathcal{B}_{\ell s}^{\mathrm{cal}}(\theta_\ell^{obs}, T_\ell^{obs}, \lambda_{\ell s})} - 1\right].$$

The numerator evaluates an implementable fixed precision, not a draw-specific optimal policy. The denominator integrates the joint posterior for the current regime.

Table 8: Contest-Level Structural Estimates and Benchmarks

Contest Id	N_i	N_j	Prize (100,000 USD)	Public Data %	Bayesian Posterior Mean						Diagnostic Benchmarks	
					λ	σ	μ_0	c_i	c_j	r	λ^{bench}	σ^{bench}
2445	15	35	0.0500	25	180.930	0.167	0.195	1.277	0.652	17.607	117.492	0.195
2454	29	21	0.0015	62	14.016	0.733	4.312	0.044	0.065	155.460	42.015	0.862
2464	19	24	0.0095	20	9.746	1.917	-0.647	0.085	0.064	118.336	9.488	2.169
2478	21	23	0.0095	30	12.557	0.448	-0.071	0.242	0.219	43.163	1445.766	1.940
2489	32	58	0.0025	10	38.426	0.477	0.428	0.101	0.052	119.881	310.279	1.292
2549	22	28	0.0050	30	5.934	0.694	-1.790	0.095	0.073	106.818	33.547	3.107
2551	115	52	0.0150	30	33.122	0.112	-0.030	0.339	0.686	43.271	6896.087	1.075
2667	97	143	0.3000	50	29.414	0.363	0.171	0.771	0.516	15.796	4.012	0.729
2749	76	90	0.0300	42	614.167	0.215	0.809	0.281	0.241	40.234	4730.298	0.604
2762	24	41	0.0100	35	5.476	0.195	0.781	0.201	0.116	65.439	3.967	2.943
2860	18	8	0.0100	30	66.251	0.176	-0.039	0.284	0.630	27.437	6436.776	0.045
3064	25	9	0.0015	25	362.219	0.090	0.007	0.112	0.343	56.168	874483.872	0.094
3080	46	52	0.0009	25	770.394	0.130	-0.466	0.076	0.069	122.299	22933.360	0.171
3288	30	20	0.0100	0	15.575	2.377	-1.218	0.064	0.102	108.953	0.090	0.803
3338	73	52	0.0150	30	303.851	0.157	0.300	0.240	0.311	43.680	692.427	0.412
3353	59	38	0.0600	30	120.357	0.796	-0.100	0.216	0.340	36.577	1791.142	0.870
3366	30	20	0.0020	50	21.455	0.168	-0.238	0.086	0.136	83.608	425.685	0.871
3507	17	18	0.0050	33	10.573	0.406	0.310	0.147	0.139	63.503	14.607	0.873
3509	15	23	0.0020	30	32.889	0.835	-0.082	0.107	0.063	107.202	40.560	0.797
3526	20	18	0.0500	20	26.414	0.172	-0.098	0.945	1.042	12.100	251.190	0.324
3641	23	110	0.0150	43	164.354	0.173	0.461	0.732	0.149	33.654	32.094	0.652
3774	126	71	0.0025	50	38.961	0.312	-1.467	0.046	0.083	135.463	54.753	1.500
3800	98	87	0.0300	15	61.911	0.520	2.896	0.126	0.143	66.733	240.703	1.697
3926	121	69	0.0030	25	1987.510	0.094	0.108	0.075	0.132	99.524	551.512	0.303
3928	26	37	0.0068	70	495.856	0.121	-0.453	0.324	0.236	39.687	5780.185	0.151
3929	143	98	0.0800	50	1.009	0.184	-0.207	0.264	0.400	26.608	898.817	6.452
3960	264	414	0.0800	40	313.566	0.477	-1.187	0.226	0.147	75.472	137.702	0.881
4104	37	19	0.2000	20	560.087	0.240	-0.030	1.056	1.660	12.278	4729.137	0.082
4366	150	141	0.0800	2	228.074	0.167	-0.258	0.254	0.305	26.776	1397.681	0.558
4407	155	276	0.0400	19	86.772	0.095	0.365	0.487	0.278	36.982	3760.994	1.226
4453	22	27	0.0200	30	18.394	0.394	1.096	0.384	0.308	29.235	1472.125	0.597
4477	11	28	0.0200	50	299.903	0.143	0.291	0.940	0.425	22.064	167.491	0.060
4488	35	8	0.0200	1	12.128	1.819	0.650	0.080	0.416	52.916	144.770	3.078
4493	46	39	0.0500	40	37.241	0.541	-0.410	0.265	0.312	34.330	460.567	2.926
4657	239	310	0.0400	30	239.856	0.070	-0.140	0.502	0.400	34.088	140654.610	1.418
4699	21	79	0.0500	30	16499.702	0.022	0.087	2.706	2.449	10.060	8603.296	0.037
4986	69	11	0.1000	50	510.876	0.178	-0.018	0.797	3.492	25.576	5782.765	0.287
5056	50	91	0.0500	33	333.495	0.274	-0.974	0.533	0.305	32.184	83025.549	0.680
5144	53	16	0.2000	20	4.442	1.585	9.925	0.240	0.805	24.730	353.818	8.205
5174	162	163	0.0300	50	106.147	0.632	-0.665	0.114	0.113	80.383	46171.708	1.435
5229	47	57	0.1500	10	12.072	2.201	0.110	0.240	0.197	43.067	14.240	2.498

Table 8 (continued)

Contest Id	N_i	N_j	Prize (100,000 USD)	Public Data %	Bayesian Posterior Mean						Diagnostic Benchmarks	
					λ	σ	μ_0	c_i	c_j	r	λ^{bench}	σ^{bench}
5261	220	176	0.1000	30	1207.612	0.502	-1.290	0.186	0.237	41.684	16160.346	0.535
5357	99	105	0.0500	30	3426.767	0.213	0.405	0.355	0.332	29.625	7672.799	0.217
5390	220	74	0.0400	30	23.603	0.133	-0.132	0.197	0.566	65.770	186.217	1.662
5497	79	40	0.0400	30	99.957	0.176	0.554	0.440	0.801	25.667	611515.682	0.571
6322	158	110	0.1000	66	6.617	0.112	-0.742	0.432	0.592	27.661	153.817	5.171
6927	80	54	0.0400	1	351.008	0.197	-0.087	0.285	0.412	32.689	1230.717	1.102
7115	63	106	0.1000	30	17.453	0.181	0.133	1.027	0.701	23.271	181616.578	4.497
7162	201	32	0.0100	50	30.009	0.893	-1.340	0.032	0.215	112.051	4115.904	2.651
7634	255	122	0.0200	2	27.596	1.290	0.903	0.045	0.095	128.184	1288.305	3.298
7878	7	9	0.0400	29	2.126	1.180	-0.535	0.659	0.510	19.692	0.961	2.073
8076	170	129	0.0600	10	76.821	0.110	-0.858	0.262	0.371	23.705	1564.803	0.988
8078	121	90	0.0400	36	1.663	1.216	2.711	0.089	0.121	84.858	28.935	7.361
8219	42	77	0.0040	22	1.755	0.424	-0.069	0.109	0.057	117.723	58937.389	4.212
8396	159	200	0.0050	1	5.837	0.905	6.780	0.047	0.037	192.620	131.304	5.108
9120	301	177	0.1000	20	555.634	0.453	0.134	0.197	0.334	42.485	5828.850	0.748
9949	48	101	0.0500	20	201.441	0.706	3.968	0.349	0.163	41.153	896.605	0.885
10200	78	77	0.0400	9	399.732	0.708	-7.317	0.176	0.181	54.897	2733.573	0.813
10684	26	113	0.0400	57	98.386	0.327	-0.349	0.780	0.180	33.652	4.966	0.800
13333	68	67	0.0200	25	14912.766	0.041	-0.077	0.483	0.540	23.220	22323.637	0.049
14242	381	118	0.0300	20	174.573	0.225	-0.017	0.120	0.390	55.703	582.054	0.759
14420	250	43	0.0800	22	13.219	1.464	-3.206	0.070	0.421	61.902	0.794	4.092
18045	103	181	0.0400	30	89.983	0.560	-2.979	0.247	0.139	54.258	11.530	1.505
19018	294	295	0.0800	30	462.296	0.678	1.173	0.155	0.156	59.240	4758.865	0.947
19991	49	244	0.0400	20	160.391	0.313	-0.101	0.726	0.146	50.210	108.909	0.848
20270	86	65	0.0400	30	97.262	0.386	-0.058	0.166	0.228	46.302	46.693	0.984
21669	173	146	0.0100	21	33.544	1.168	-0.983	0.040	0.048	176.541	400.378	2.684
22962	99	61	0.0400	24	157.642	0.712	-0.856	0.124	0.204	59.076	2633.166	0.910
23249	58	145	0.0100	16	17908.283	0.031	-0.026	0.966	0.710	56.674	249344.779	0.081
23652	85	81	0.0100	20	92.972	2.112	-0.160	0.040	0.042	197.386	11873.949	1.658
37077	33	199	0.0400	24	301.674	0.842	0.663	0.505	0.081	54.044	581.433	0.904
38128	48	152	0.0800	42	353.804	0.109	-0.253	3.292	1.263	37.200	1624.612	0.630
38760	13	50	0.0500	33	2.842	1.643	-0.068	0.432	0.102	47.236	41507.819	5.596

C.2. Focal-Pair and Score-Scale Robustness

The alternative-pair exercise ranks contestants by their last recorded private score, using submission-ID order to resolve records on the same date and contestant ID to break score ties. It excludes the baseline contestant i and searches the remaining candidates within the top 20 in rank order. This ranking need not coincide with the final leaderboard based on contestants' best scores. Each contestant must have an activity span of at least ten days, the pair must overlap for more than ten days, and each must contribute at least five submissions to the retained count window. Prize conventions, the model horizon, and the within-day time-imputation rule are retained from the baseline. The initialization prefix is recomputed from the alternative pair's first

jointly available scores, changing the count-window start in 43 of 73 contests. Thus, the exercise compares alternative pairs with their corresponding usable data, rather than holding the effective observation window fixed.

Score-scale calibration uses submitted public scores from the baseline focal pairs, without private scores or repeated carried-forward values. We center their logit-transformed levels by each contest's median for calibration only, assign equal total weight to each contest, and compute pooled weighted quantiles q_j at $p_j = (j - \frac{1}{2})/999$, $j = 1, \ldots, 999$. The L_1 choice minimizes $\sum_j |aq_j - \Phi^{-1}(p_j)|$ over $a > 0$; the L_2 choice minimizes $\sum_j [aq_j - \Phi^{-1}(p_j)]^2$. They give $a = 1.444$ and $a = 0.781$, respectively. Public gaps are multiplied by the selected a for estimation, and private gaps by the same a for diagnostics; the calibration centering does not enter those gaps.

All 219 robustness fits (73 alternative-pair fits and 73 fits at each score scale) have no divergent transitions or maximum-tree-depth saturations, with maximum split $\widehat{R} = 1.009$. Re-estimation retains the baseline numerical priors and parameter bounds in the transformed units, so this exercise includes sensitivity to their scale-dependent regularization. Table 9 reports the benchmark regressions.

Table 9 Private-Leaderboard Benchmark Regressions under Robustness Specifications

Specification	Slope	Standard error	95% confidence interval	R^2
Signal precision λ				
Baseline ($a = 1$)	0.717	0.156	[0.412, 1.022]	0.218
Alternative pair	0.638	0.159	[0.326, 0.950]	0.148
L_1 scale	0.712	0.154	[0.410, 1.015]	0.217
L_2 scale	0.725	0.154	[0.423, 1.027]	0.221
Innovation risk σ				
Baseline ($a = 1$)	0.937	0.209	[0.526, 1.347]	0.282
Alternative pair	1.309	0.165	[0.985, 1.633]	0.601
L_1 scale	0.862	0.189	[0.492, 1.232]	0.276
L_2 scale	0.996	0.226	[0.553, 1.439]	0.289

Note: Each regression uses 73 contests and an intercept. The dependent variable is $\log(1 + \lambda^{\text{bench}})$ or $\log(1 + \sigma^{\text{bench}})$; the regressor is the corresponding log-transformed posterior mean estimate. Estimates and benchmarks use each specification's score units. All slopes have $p < 0.001$. Standard errors and confidence intervals use HC1 heteroskedasticity-robust covariance estimates.

For comparisons in common score units, signal precision is $a^2\hat{\lambda}_a$ and innovation risk is $\hat{\sigma}_a/a$, where the subscript a denotes the re-estimated scale. Table 10 compares their cross-contest rankings with the baseline and repeats the posterior-average precision optimization from Section 5.3. Full-disclosure versus finite-precision classifications agree with the baseline for all 73 contests under both scales. At the 75% posterior-support threshold, the numbers supporting increased precision, reduced precision, or neither direction are $(67, 3, 3)$ in the baseline, $(66, 5, 2)$ under L_1, and $(67, 1, 5)$ under L_2.

Table 10 Score-Scale Sensitivity of Parameter Rankings and Precision Design

Specification	a	Rank correlation λ; σ	Full disclosure	Finite precision	Median relative output gain (%)
Baseline	1.000	—	67	6	0.156
L_1	1.444	0.995; 0.993	67	6	0.109
L_2	0.781	0.998; 0.995	67	6	0.166

Note: Each specification covers the same 73 contests. Rank correlations are Spearman correlations between common-unit posterior means and baseline posterior means. Precision recommendations maximize posterior mean calibrated output at the observed prize and horizon. Output gains compare one recommended precision per contest with its current regime, integrating the joint posterior as in Table 7. The median summarizes variation across contests, not posterior uncertainty.

Acknowledgments

The authors gratefully acknowledge the valuable research assistance provided by Simian Zhang during the course of this study.

References

Ales L, Cho SH, Körpeoğlu E (2017) Optimal award scheme in innovation tournaments. Operations Research 65(3):693–702, URL http://dx.doi.org/10.1287/opre.2016.1575.

Ales L, Cho SH, Körpeoğlu E (2021) Innovation tournaments with multiple contributors. Production and Operations Management 30(6):1772–1784, URL http://dx.doi.org/10.1111/poms.13342.

Aoyagi M (2010) Information feedback in a dynamic tournament. Games and Economic Behavior 70(2):242–260, URL http://dx.doi.org/10.1016/j.geb.2010.01.013.

Barrau A, Bonnabel S (2017) The invariant extended kalman filter as a stable observer. IEEE Transactions on Automatic Control 62(4):1797–1812, ISSN 0018-9286, 1558-2523, URL http://dx.doi.org/10.1109/TAC.2016.2594085.

Bensoussan A (1992) Stochastic Control of Partially Observed Systems (Cambridge University Press), 1 edition, ISBN 9780511526503, URL http://dx.doi.org/10.1017/CBO9780511526503.

Berge C (1963) Topological Spaces: Including a Treatment of Multi-valued Functions, Vector Spaces and Convexity (Edinburgh and London: Oliver and Boyd).

Bergemann D, Heumann T, Morris S, Sorokin C, Winter E (2022) Optimal information disclosure in classic auctions. American Economic Review: Insights 4(3):371–388, ISSN 2640-205X, 2640-2068, URL http://dx.doi.org/10.1257/aeri.20210504.

Bimpikis K, Ehsani S, Mostagir M (2019) Designing dynamic contests. Operations Research 67(2):339–356, ISSN 0030-364X, 1526-5463, URL http://dx.doi.org/10.1287/opre.2018.1823.

Budd C, Harris C, Vickers J (1993) A model of the evolution of duopoly: Does the asymmetry between firms tend to increase or decrease? The Review of Economic Studies 60(3):543–573, URL http://dx.doi.org/10.2307/2298124.

Frogerais P, Bellanger JJ, Senhadji L (2012) Various ways to compute the continuous-discrete extended kalman filter. IEEE Transactions on Automatic Control 57(4):1000–1004, ISSN 0018-9286, 1558-2523, URL http://dx.doi.org/10.1109/TAC.2011.2168129.

Gronau QF, Singmann H, Wagenmakers EJ (2020) bridgesampling: An R package for estimating normalizing constants. Journal of Statistical Software 92(10), URL http://dx.doi.org/10.18637/jss.v092.i10.

Guerre E, Perrigne I, Vuong Q (2000) Optimal nonparametric estimation of first-price auctions. Econometrica 68(3):525–574, ISSN 0012-9682, 1468-0262, URL http://dx.doi.org/10.1111/1468-0262.00123.

Harris C, Vickers J (1987) Racing with uncertainty. The Review of Economic Studies 54(1):1–21, ISSN 00346527, 1467937X, URL http://www.jstor.org/stable/2297442.

Hoffman MD, Gelman A (2014) The no-u-turn sampler: Adaptively setting path lengths in hamiltonian monte carlo. Journal of Machine Learning Research 15(47):1593–1623, URL http://jmlr.org/papers/v15/hoffman14a.html.

Huang PC, Zhang Z, Erat S (2024) Designing Data Science Contests: The Role of Training vs Test Splits. URL http://dx.doi.org/10.2139/ssrn.4976827.

Huang Y, He M (2021) Structural analysis of tullock contests with an application to U.S. house of representatives elections. International Economic Review 62(3):1011–1054, ISSN 1468-2354, URL http://dx.doi.org/10.1111/iere.12502.

Hwang SH (2012) Technology of military conflict, military spending, and war. Journal of Public Economics 96(1-2):226–236, ISSN 00472727, URL http://dx.doi.org/10.1016/j.jpubeco.2011.09.003.

Jackson J (2024) On quasilinear parabolic systems and FBSDEs of quadratic growth. The Annals of Applied Probability 34(1A):357–387, URL http://dx.doi.org/10.1214/23-AAP1966.

Jiang ZZ, Huang Y, Beil DR (2022) The role of feedback in dynamic crowdsourcing contests: A structural empirical analysis. Management Science 68(7):4858–4877, ISSN 0025-1909, 1526-5501, URL http://dx.doi.org/10.1287/mnsc.2021.4140.

Kang K (2016) Policy influence and private returns from lobbying in the energy sector. The Review of Economic Studies 83(1):269–305, ISSN 0034-6527, 1467-937X, URL http://dx.doi.org/10.1093/restud/rdv029.

Kato T (1980) Perturbation Theory for Linear Operators (Berlin: Springer-Verlag), 2 edition, corrected printing.

Khorasani S, Nittala L, Krishnan VV (2023) Screening in multistage contests. Manufacturing & Service Operations Management msom.2021.0378, ISSN 1523-4614, 1526-5498, URL http://dx.doi.org/10.1287/msom.2021.0378.

Lemus J, Marshall G (2021) Dynamic tournament design: Evidence from prediction contests. Journal of Political Economy 129(2):383–420, ISSN 0022-3808, URL http://dx.doi.org/10.1086/711762.

Lewis PAW, Shedler GS (1979) Simulation of nonhomogeneous poisson processes by thinning. Naval Research Logistics Quarterly 26(3):403–413, ISSN 0028-1441, 1931-9193, URL http://dx.doi.org/10.1002/nav.3800260304.

Lorenzi L, Lunardi A, Metafune G, Pallara D (2005) Analytic semigroups and reaction-diffusion problems. URL https://people.dmi.unipr.it/alessandra.lunardi/LectureNotes/I-Sem2005.pdf, internet Seminar 2004–2005.

Mihm J, Schlapp J (2019) Sourcing innovation: On feedback in contests. Management science 65(2):559–576, URL http://dx.doi.org/10.1287/mnsc.2017.2955.

Moldovanu B, Sela A (2001) The Optimal Allocation of Prizes in Contests. American Economic Review 91(3):542–558, ISSN 0002-8282, URL http://dx.doi.org/10.1257/aer.91.3.542.

Moscarini G, Smith L (2011) Optimal dynamic contests, URL https://campuspress.yale.edu/moscarini/files/2017/01/contests_current-x5l31k.pdf, working paper.

Nishida T (1978) Nonlinear Hyperbolic Equations and Related Topics in Fluid Dynamics. Number 78-02 in Publications Mathématiques d'Orsay (Orsay: Université de Paris-Sud, Département de Mathématique).

Risdal M, Bozsolik T (2022) Meta kaggle. URL http://dx.doi.org/10.34740/KAGGLE/DS/9.

Ryvkin D (2022) To fight or to give up? dynamic contests with a deadline. Management Science 68(11):8144–8165, ISSN 0025-1909, 1526-5501, URL http://dx.doi.org/10.1287/mnsc.2021.4206.

Shakhgildyan K (2022) Nonparametric identification and estimation of all-pay auction and contest models. Review of Economic Design ISSN 1434-4750, URL http://dx.doi.org/10.1007/s10058-022-00309-3.

Skaperdas S (1996) Contest success functions. Economic Theory 7(2):283–290, ISSN 0938-2259, 1432-0479, URL http://dx.doi.org/10.1007/BF01213906.

Yoganarasimhan H (2016) Estimation of beauty contest auctions. Marketing Science 35(1):27–54, ISSN 0732-2399, URL http://dx.doi.org/10.1287/mksc.2015.0929.

Zhang S, Singh PV, Ghose A (2019) A structural analysis of the role of superstars in crowdsourcing contests. Information Systems Research 30(1):15–33, ISSN 1047-7047, URL http://dx.doi.org/10.1287/isre.2017.0767.